

Feasibility of using the ADS1299

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Objective

- Investigate the feasibility of acquiring and processing electromyography (EMG) signals using the ADS1299 analog front-end and MSPM0C1104 microcontroller.
- Specifically:
 1. Understand key characteristics of EMG signals (frequency range, amplitude, bandwidth, etc.).
 2. Evaluate if the ADS1299 is suitable for EMG signal acquisition.
 3. Design a connection and data flow plan between ADS1299 and MSPM0C1104.

Methods

Literature Review:

- Analyzed Rashid et al. [1] and Li et al. [2] to assess the ADS1299's performance in acquiring biopotential signals under experimental conditions.

Chip Evaluation Approach:

- Extracted and analyzed ADS1299 specifications (gain, noise, sampling rate) from the TI datasheet.

Interface Design Methodology:

- Designed a high-level connection diagram based on ADS1299 and MSPM0C1104 documentation.

Results

Rashit et al. [1] demonstrated that the ADS1299, and 8-channel, 24-bit ADC, can record biopotentials with laboratory grade precision.

Quote: “The main findings of this research are that the ADS1299 is comparable with respect to power across EEG bands, power ratio, pre-movement noise of the EEG, and the signal-to-noise ratio, the amplitude and time of the negative peak, and cosine similarity of the MRCPs” [1].

In Rashid et al.'s [1] work, MRCPs are used as a benchmark signal to evaluate the quality of EEG acquisition by the ADS1299 compared to a laboratory-grade amplifier. Because MRCPs are:

- Low-frequency (~ 0.1 – 5 Hz)
- Very low amplitude (5 – 50 μV)
- Time-locked to movement

ADC1299 Specs:

- Input-referred noise: ~ 1 $\mu\text{V RMS}$
- Programmable gain: $1\times$ – $24\times$
- Sampling range: 250 SPS – 16 kSPS
- Digital filter: Built-in decimation filter provides strong low-pass response.

Note: This assumes that due to EEG being a harder signal to capture this would imply that the ADS is more than sufficient for EMG.

Results

- Li et al. [2] designed and implemented an 8-channel surface EMG (sEMG) acquisition system using the ADS1299 analog front-end.
- The system integrates a high-density electrode array, bias-drive circuitry, and an STM32 microcontroller for signal processing.
- They focused on back-of-hand EMG acquisition during voluntary fist-clenching movements.
- Provides direct experimental proof that the ADS1299 is effective for EMG acquisition, not just EEG [2].

Connecting ADS1299 to MSPM0C1104

- Communication: SPI bus used for configuration and data readout, as described in [1].
- Signal flow:
 1. MSPM0C1104 configures ADS1299 registers over SPI.
 2. ADS1299 digitizes EMG signals and signals data readiness (GPIO).
 3. MSPM0C1104 retrieves and processes 24-bit samples.

Example Layout

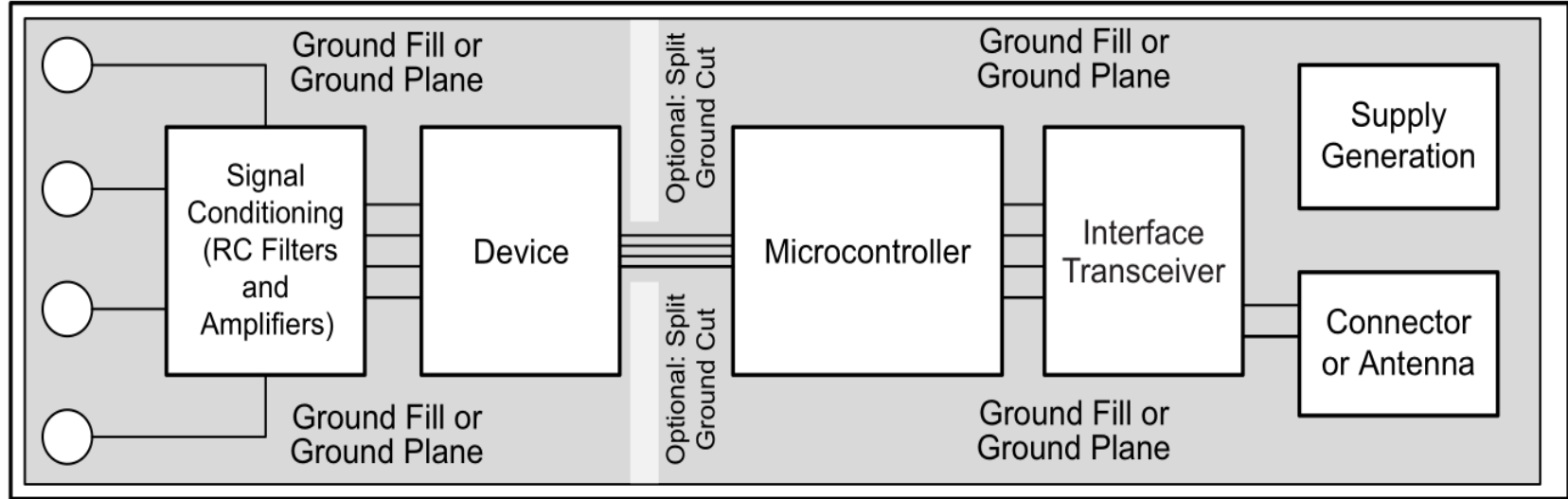


Figure 79. System Component Placement

Example Application

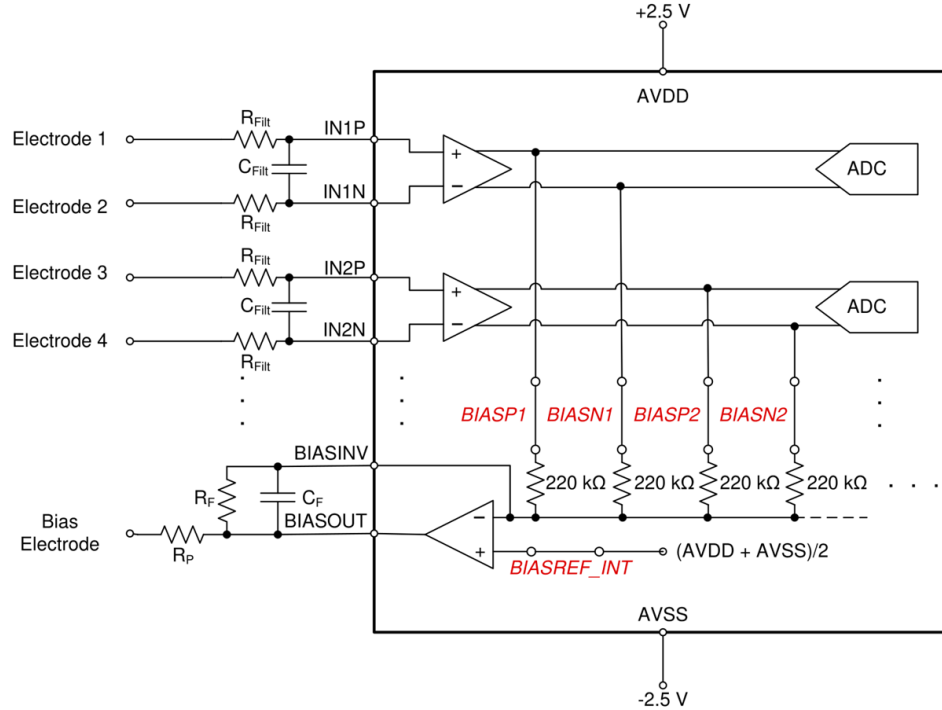


Figure 72. Example Schematic Using the ADS1299 in an EEG Data Acquisition Application, Sequential Montage

Observation/Conclusion:

- Options for the proof of concept:
 - ADS1299 PDK (Performance Demonstration Kit)
 - Fast, but potentially harder to integrate with main circuit
 - ADS1299 on a separate PCB
 - Need to properly design first, but would be easier to integrate with the main PCB

EMG Signals (Questions to ask)

- What level of noise should we expect in real EMG measurements, and how much filtering is typically required?
- Should we consider active filtering (analog) before the ADS1299, or is digital filtering sufficient post-acquisition?
- Record EMG as in store the signal data in a file or just read the EMG signals and provide it as an output?
 - For how long and how much?

Next steps:

Determine the option for the proof of concept

Help Jeremiah plan how integrate the ADS1299 into the current design.

Reference

- [1] U. Rashid, I. Niazi, N. Signal, and D. Taylor, "An EEG experimental study evaluating the performance of Texas Instruments ADS1299," *Frontiers in Neuroscience*, vol. 12, p. 627, 2018, doi: 10.3390/s18113721
- [2] Y. Li, H. Pan, and Q. Song, "ADS1299-Based Array Surface Electromyography Signal Acquisition System," *J. Phys.: Conf. Ser.*, vol. 2383, 012054, 2022. [Online]. Available: <https://doi.org/10.1088/1742-6596/2383/1/012054>

<https://www.ti.com/product/ADS1299>

<https://www.ti.com/tool/ADS1299EEGFE-PDK>

Initial Power Amplifier Design

Luis Wong 10/21/2025

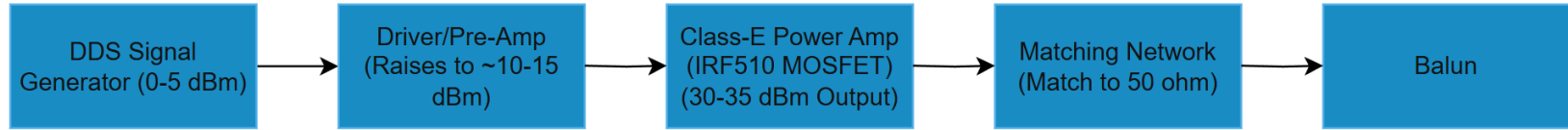
Objective

- To fully understand and attempt design an RF Power Amplifier for use in device.
 - We need 30 - 35 dBm at output, DDS can give 0-5 dBm.
 - 50-ohm resistance

Methods

- Read through various sources to better understand the basics of RF PAs due to being unfamiliar with the subject.
- Model a Class-E power amplifier to efficiently convert low-power DDS output into the required output power.
 - Will make use of PA used in TI Reference Design Guide [1] and also designs found in [3],[5] as a guide.

Results



High-Level View of the PAs design.

- **Power Switch (MP5000ADQ-LF-P):** Disconnects PA stage when not transmitting.

Results

- Different classes of power amplifiers [2],[6],[7]:

Class	Conduction Angle	Power Efficiency	Linearity
A	360° (always on)	~25-35%	Highest
B	180°	~65%	Moderate
AB	180-360°	~50-70%	Better than AB
C	>360°	~60-80%	Poor
D	Switch	~80-90%	Poor
E	Switch	~80-90%	Moderate
F	Switch + harmonic tuning	~85-90%	Moderate

Results



Power to Voltage Conversion Table

P (dBm)	P (mW)	V _{max} (V)	V _p (V) ¹	V _{pp} (V)	P (dBm)	P (mW)	V _{max} (V)	V _p (V) ¹	V _{pp} (V)
-30	0.001	0.007	0.010	0.020	0	1.000	0.224	0.316	0.632
-29	0.001	0.008	0.011	0.022	1	1.259	0.251	0.355	0.710
-28	0.002	0.009	0.013	0.025	2	1.585	0.282	0.398	0.796
-27	0.002	0.010	0.014	0.028	3	1.995	0.316	0.447	0.891
-26	0.003	0.011	0.016	0.032	4	2.512	0.354	0.501	1.002
-25	0.003	0.013	0.018	0.036	5	3.162	0.398	0.562	1.125
-24	0.004	0.014	0.020	0.040	6	3.981	0.446	0.631	1.262
-23	0.005	0.016	0.022	0.045	7	5.012	0.501	0.708	1.416
-22	0.006	0.018	0.025	0.050	8	6.310	0.562	0.794	1.589
-21	0.008	0.020	0.028	0.056	9	7.943	0.630	0.891	1.783
-20	0.010	0.022	0.032	0.063	10	10.000	0.707	1.000	2.000
-19	0.013	0.025	0.035	0.071	11	12.589	0.793	1.122	2.244
-18	0.016	0.028	0.040	0.080	12	15.849	0.890	1.259	2.512
-17	0.020	0.032	0.045	0.089	13	19.953	0.999	1.413	2.825
-16	0.025	0.035	0.050	0.100	14	25.119	1.121	1.585	3.170
-15	0.032	0.040	0.056	0.112	15	31.623	1.257	1.778	3.557
-14	0.040	0.045	0.063	0.126	16	39.811	1.411	1.995	3.991
-13	0.050	0.050	0.071	0.142	17	50.119	1.583	2.239	4.477
-12	0.063	0.056	0.079	0.159	18	63.096	1.776	2.512	5.024
-11	0.079	0.063	0.089	0.178	19	79.433	1.993	2.818	5.637
-10	0.100	0.071	0.100	0.200	20	100.000	2.236	3.162	6.325
-9	0.126	0.079	0.112	0.224	21	125.893	2.509	3.548	7.096
-8	0.158	0.089	0.126	0.252	22	158.489	2.815	3.981	7.962
-7	0.200	0.100	0.141	0.283	23	199.526	3.159	4.467	8.934
-6	0.251	0.112	0.158	0.317	24	251.189	3.544	5.012	10.024
-5	0.316	0.126	0.178	0.356	25	316.228	3.976	5.623	11.247
-4	0.398	0.141	0.200	0.399	26	398.107	4.462	6.310	12.619
-3	0.501	0.158	0.224	0.448	27	501.187	5.006	7.079	14.129
-2	0.631	0.178	0.251	0.502	28	630.957	5.617	7.943	15.887
-1	0.794	0.199	0.282	0.564	29	794.328	6.302	8.913	17.825
0	1.000	0.224	0.316	0.632	30	1000.000	7.071	10.000	20.000

¹For Square wave signal, $V_p = 1/2 V_{max}$.
Note: The converted voltages in the above table are for $R = 50 \Omega$

$$P_{\text{dBm}} = 10 \log_{10} P_{\text{(mW)}}$$

$$P_{\text{(mW)}} = 10^{(P_{\text{dBm}}/10)}$$

$$P_{\text{(mW)}} = (V_{\text{max}}/R)^2 \cdot 10^3/R$$

$$V_{\text{max}}(\text{V}) = \sqrt{(P_{\text{(mW)}} \cdot R / 10^3)}$$

$$V_p = \sqrt{2} \cdot V_{\text{max}} \text{ for sinusoid - Fig. a.}$$

$$V_p = V_{\text{max}} \text{ for square wave - Fig. b.}$$

$$V_{p-p} = 2 \cdot V_p$$

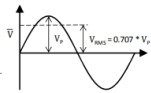


Fig. a. Sinusoidal signal

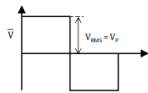


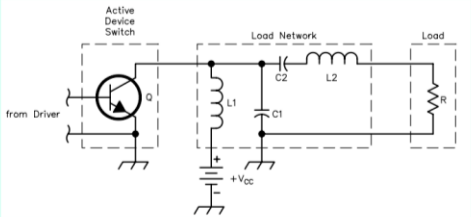
Fig. b. Square wave signal

<https://markimicrowave.com/tools/power-to-voltage.pdf>

Results

Enter Vcc: 12 V
 Enter Vo (transistor FET saturation voltage): 0 V
 Enter Coss (transistor FET output capacitance): 10p F
 Enter frequency: 25M Hz
 Enter the desired power output: 1 W
 Enter L1: 50u H
 Calculate

R is: 74.39 Ω (Impedance transformer to 50Ω: secondary (50Ω) to primary (amplifier) turns ratio: 819.81 m)
 C1 is: 8.4 p F (XC1 is: 757.78 Ω)
 C2 is: 22.86 p F (XC2 is: 278.46 Ω)
 L1min is: 41.27 μ Ω (XL1 is: 7.85 k Ω)
 L2 is: 2.37 μ H (XL2 is: 371.97 Ω)
 XL2 - XC2 = 93.52 Ω
 Expect Icc of at least: 83.33 mA
 Select a FET transistor with a max Vds/Vce of at least: 63.4 V (Includes a safety factor of 0.8)
 1 / f1 = 21.63 M Hz. When Q1 closed, should be less than the frequency of operation.
 1 / f2 = 32.39 M Hz. When Q1 open, should be greater than the frequency of operation.
 1 / (f1 + f2 half period) = 27.01 M Hz. Should be about equal to the frequency of operation.



Schematic from Nathan Sokal's paper titled "Class-E RF power amplifiers", published in QEX Jan/Feb 2001.

For more info on designing Class-E amplifiers for LF/MF, see [YKISV's Class-E for beginners](#)
 L1min and Fr1/2 calculations are from [here](#).
 Many thanks to Daniel O'Connor (doconnor AT gsoft.com.au) for his contribution of the SI prefix capability, the L1min and Fr1/2 calculation as well as a general tidy up of the javascript code!
 Back to [calculators page](#) or [main page](#).

$$R = \left(\frac{V_{CC} - V_o}{P} \right)^2 0.576801 \left(1.0000086 - \frac{0.414395}{Q_L} - \frac{0.577501}{Q_L^2} + \frac{0.205967}{Q_L^3} \right)$$

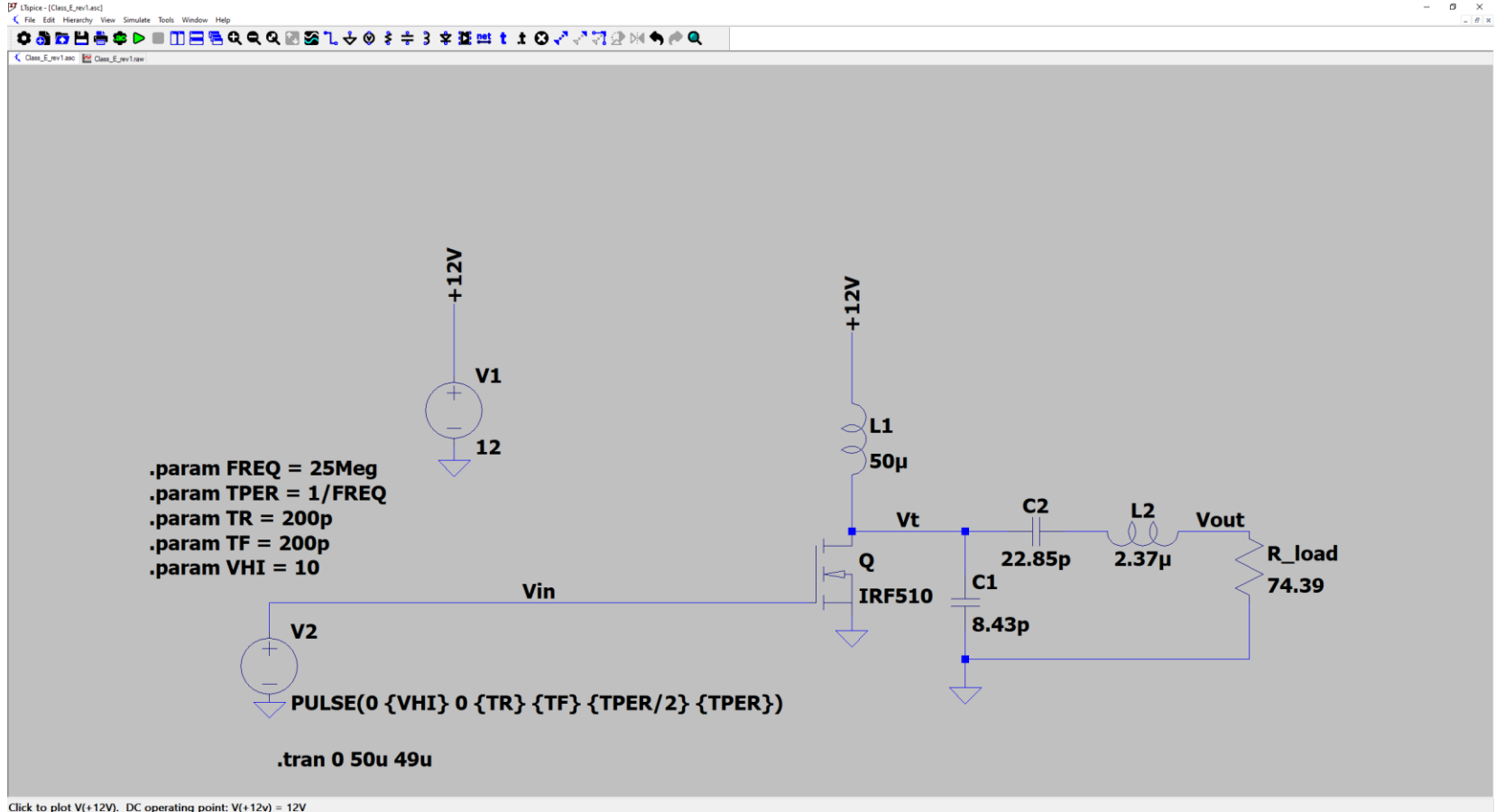
$$C1 = \frac{1}{34.2219 f R} \left(0.99866 + \frac{0.91424}{Q_L} - \frac{1.03175}{Q_L^2} \right) + \frac{0.6}{(2\pi f)^2 L1}$$

$$C2 = \frac{1}{2\pi f R} \left(\frac{1}{Q_L - 0.104823} \right) \left(1.00121 + \frac{1.01468}{Q_L - 1.7879} \right) - \frac{0.2}{(2\pi f)^2 L1}$$

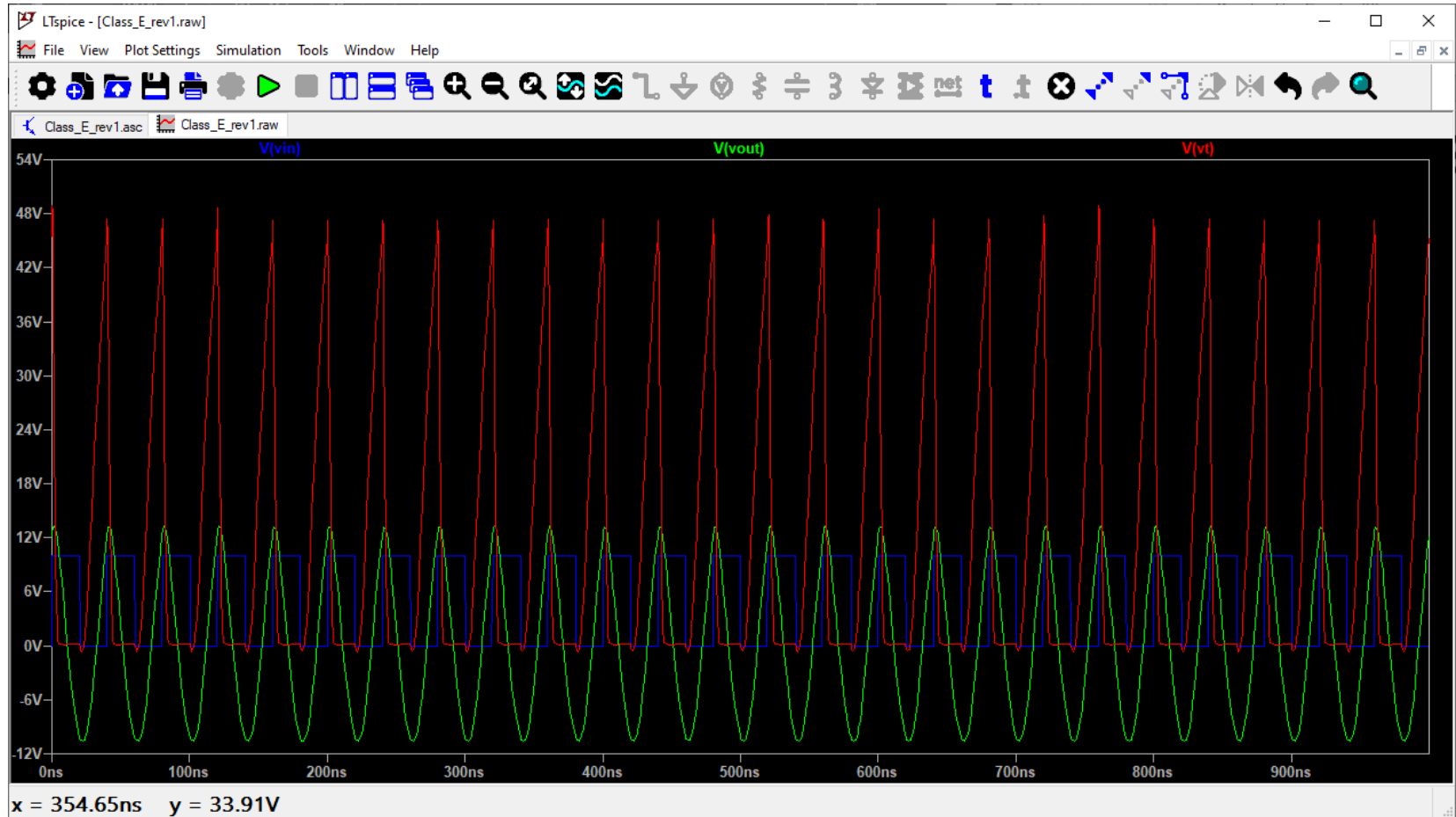
$$L2 = \frac{Q_L R}{2\pi f}$$

<https://people.physics.anu.edu.au/~dxt103/calculators/class-e.php>

Results



Results



Observation/Conclusion

- Class-E PAs provide the best trade-off between high efficiency ($\sim 80\text{--}90\%$), simplicity, and output power for powering of implants [1], [3], [4], [7].
- The current Class-E PA isn't designed to see $50\ \Omega$ directly so we could use:
 - Use a transformer with $\sim 1.22:1$ (from the calculator) voltage ratio (secondary to primary)
 - Use a simple L-network to transform $50\ \Omega \rightarrow 74.39\ \Omega$ instead of a physical transformer.
- For wideband operation, push-pull Class-AB could be explored.

Next Steps

Refine design and integrate it into the overall device with the help of Jerimiah through the use of Altium.

References

- [1] N. R., V. A., B. Chokkalingam, S. Padmanaban, and Z. Leonowicz, "Class E Power Amplifier Design and Optimization for the Capacitive Coupled Wireless Power Transfer System in Biomedical Implants," *Energies*, vol. 10, no. 9, p. 1409, Sep. 2017, doi: <https://doi.org/10.3390/en10091409>.
- [2] M. Ludens, "RF Power Amplifier Design," ludens.cl, 2020. <https://ludens.cl/Electron/RFamps/RFamps.html>
- [3] Y. Ben Fadhel, S. Ktata, K. Sedraoui, S. Rahmani, and K. Al-Haddad, "A Modified Wireless Power Transfer System for Medical Implants," *Energies*, vol. 12, no. 10, p. 1890, May 2019, doi: <https://doi.org/10.3390/en12101890>.
- [4] S. R. Khan, S. K. Pavuluri, G. Cummins, and M. P. Y. Desmulliez, "Wireless Power Transfer Techniques for Implantable Medical Devices: a Review," *Sensors*, vol. 20, no. 12, p. 3487, Jun. 2020, doi: <https://doi.org/10.3390/s20123487>.
- [5] Texas Instruments, "HF Power Amplifier (Reference Design Guide) RFID Systems / ASP 1.) Scope," 2004. Accessed: Oct. 21, 2025. [Online]. Available: https://e2e.ti.com/cfs-file/__key/telligent-evolution-components-attachments/00-667-00-00-00-11-91-53/appnote_5F00_trf796x_5F00_pwramp_5F00_4w.pdf
- [6] S. C. Cripps, *RF Power Amplifiers for Wireless Communications*, 2nd ed., Norwood, MA, USA: Artech House, 2006.
- [7] F. H. Raab *et al.*, "Power amplifiers and transmitters for RF and microwave," *IEEE Transactions on Microwave Theory and Techniques*, vol. 50, no. 3, pp. 814–826, Mar. 2002. doi: 10.1109/22.989965.

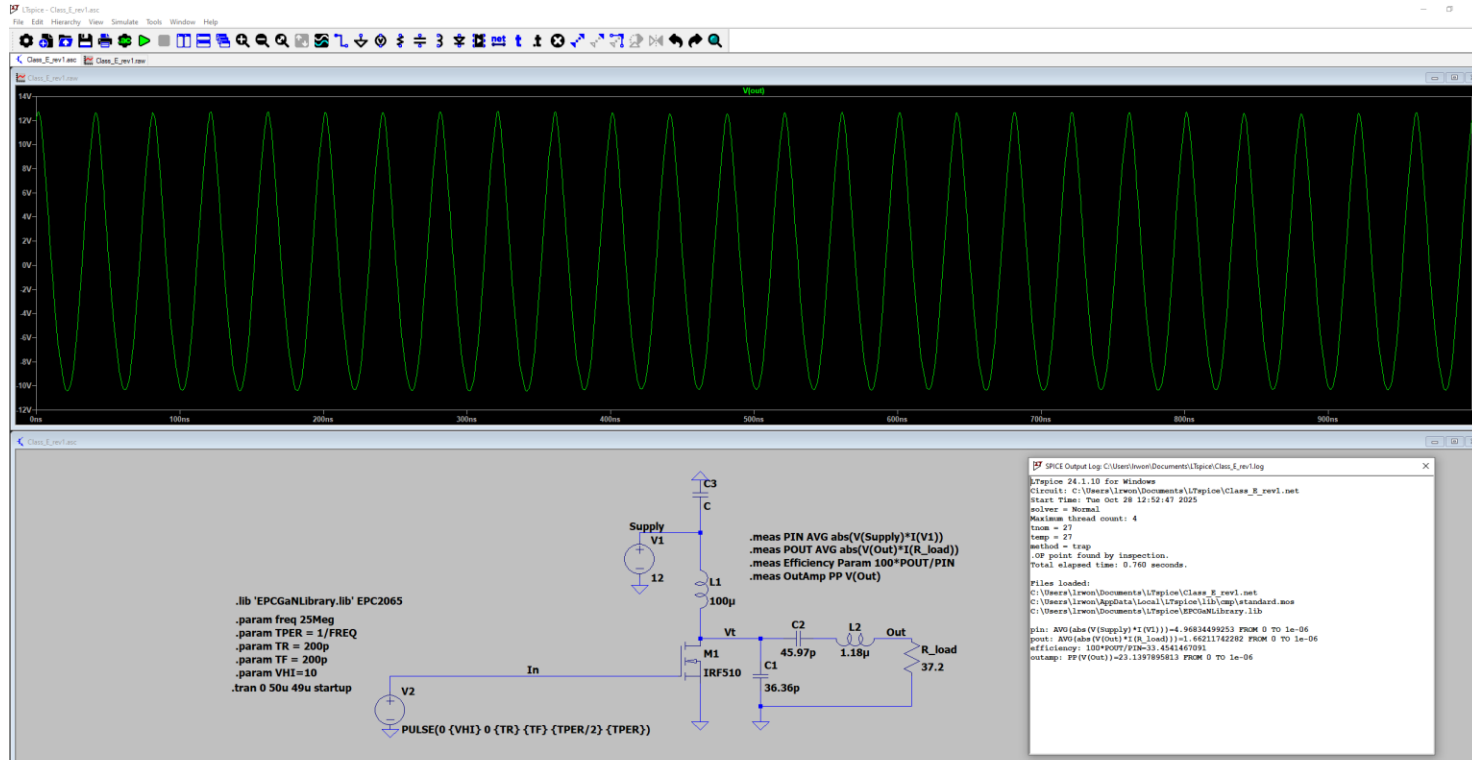
Power Amplifier Design Continued...

Luis Wong 10/28/2025

Methods

- Made use of LTspice to graph different aspects of the design and modify accordingly.
 - Compared power efficiency from 20-25MHz
- Some further research on Class E implementations.
- Used Altium in order to make schematic (at the request of Jerimiah).

Results



Results

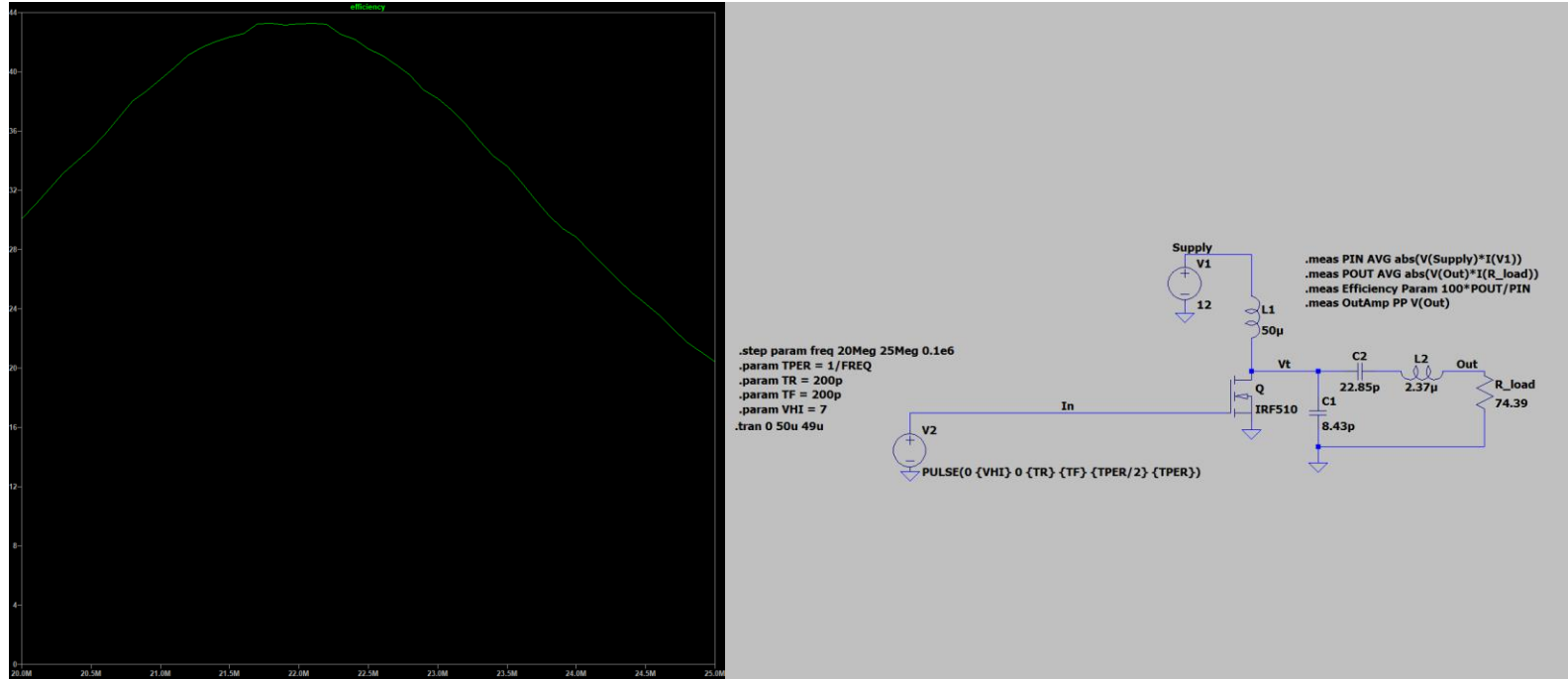


Figure 1: Efficiency graph of IRF510 (left) made using the following schematic (right) and calculated as described in the spice directives on the top-right of the design.

Results

Dynamic						
Input capacitance	C_{iss}	$V_{GS} = 0\text{ V},$ $V_{DS} = 25\text{ V},$ $f = 1.0\text{ MHz, see fig. 5}$		-	180	-
Output capacitance	C_{oss}			-	81	-
Reverse transfer capacitance	C_{rss}			-	15	-
Total gate charge	Q_g	$V_{GS} = 10\text{ V}$	$I_D = 5.6\text{ A}, V_{DS} = 80\text{ V}$ $V_{DS} = 10\text{ V},$ see fig. 6 and fig. 13 ^b	-	-	8.3
Gate-source charge	Q_{gs}			-	-	2.3
Gate-drain charge	Q_{gd}			-	-	3.8

Figure 2: From Datasheet for IRF510. Shows C_{oss} of 81pF

- If the calculations result in a shunt capacitance lower than C_{oss} this would eliminate the need of the capacitor entirely.
- Adding any external capacitance would further slow down the drain transition and ruin ZVS.
- The device is too “capacitively heavy” for the target frequency and load.
- Will lead to the lower than expected efficiency (expecting ~90%).

Results

- Redid calculations with Coss in mind.

Enter Q (around 5 is a good start):

Enter Vcc: V

Enter Vo (transistor/FET saturation voltage): V

Enter Coss (transistor/FET output capacitance): F

Enter frequency: Hz

Enter the desired power output: W

Enter L1: H

R is: Ω (Impedance transformer to 50 Ω : secondary (50 Ω) to primary (amplifier) turns ratio:

C1 is: F (XC1 is: Ω)

C2 is: F (XC2 is: Ω)

L1min is: Ω (XL1 is: Ω)

L2 is: H (XL2 is: Ω)

XL2 - XC2 = Ω

Expect Icc of at least: A

Select a FET/transistor with a max Vds/Vce of at least: V (Includes a safety factor of 0.8)

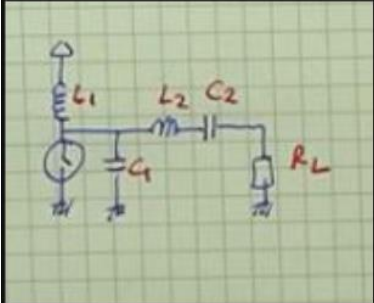
1 / fr1 = Hz. When Q1 closed, should be less than the frequency of operation.

1 / fr2 = Hz. When Q1 open, should be greater than the frequency of operation.

1 / (fr1 + f2 half period) = Hz. Should be about equal to the frequency of operation.

Figure 3:

<https://people.physics.anu.edu.au/~dxt103/calculators/class-e.php>



```
V_supply = 12;  
Pout = 1;  
Zout = 50;  
f = 25*10^7;  
Q = 5; %ranges from 0-10  
w = 2*pi*f;  
Coss = 9.5e-12;  
  
Coss = 9.5000e-12  
  
R_L = 1.154*((V_supply^2)/(2*Pout))  
  
R_L = 83.0880
```

Figure 4: R_L calculation based on this video:

<https://www.youtube.com/watch?v=Tgrakttus3c&t=1s>

Results

Dynamic Characteristics [#] (T _j = 25°C unless otherwise stated)						
PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
C _{ISS}	Input Capacitance	V _{GS} = 0 V, V _{DS} = 50 V		14	17	pF
C _{RSS}	Reverse Transfer Capacitance			0.1		
C _{OSS}	Output Capacitance			6.5	10	
C _{OSS(ER)}	Effective Output Capacitance, Energy Related (Note 2)	V _{GS} = 0 V, V _{DS} = 0 to 50 V		9.5		
C _{OSS(TR)}	Effective Output Capacitance, Time Related (Note 3)			12		
R _G	Gate Resistance			0.5		Ω
Q _G	Total Gate Charge	V _{GS} = 5 V, V _{DS} = 50 V, I _D = 0.1 A		115	145	pC
Q _{GS}	Gate-to-Source Charge	V _{DS} = 50 V, I _D = 0.1 A		32		
Q _{GD}	Gate-to-Drain Charge			25		
Q _{G(TH)}	Gate Charge at Threshold			24		
Q _{OSS}	Output Charge	V _{GS} = 0 V, V _{DS} = 50 V		600	900	
Q _{RR}	Source-Drain Recovery Charge			0		

[#] Defined by design. Not subject to production test.

All measurements were done with substrate connected to source.

Note 2: C_{OSS(ER)} is a fixed capacitance that gives the same stored energy as C_{OSS} while V_{DS} is rising from 0 to 50% BV_{DSS}.

Note 3: C_{OSS(TR)} is a fixed capacitance that gives the same charging time as C_{OSS} while V_{DS} is rising from 0 to 50% BV_{DSS}.

Figure 5: EPC2037 datasheet showing 9.5pF Coss. Might make this better for our purposes.

EPC2037 – Enhancement Mode Power Transistor

 V_{DS} , 100 V $R_{DS(on)}$, 550 mΩ I_D , 1.7 A

Revised August 21, 2024

Gallium Nitride's exceptionally high electron mobility and low temperature coefficient allows very low $R_{DS(on)}$ while its lateral device structure and majority carrier diode provide exceptionally low Q_g and zero Q_{rr} . The end result is a device that can handle tasks where very high switching frequency, and low on-time are beneficial as well as those where on-state losses dominate.

Application Notes:

- Easy-to-use and reliable gate, Gate Drive ON = 5 V typical, OFF = 0 V (negative voltage not needed)
- Top of FET is electrically connected to source

Questions:
Ask a GaN
Expert



Maximum Ratings			
PARAMETER		VALUE	UNIT
V_{DS}	Drain-to-Source Voltage (Continuous)	100	V
	Drain-to-Source Voltage (up to 10,000 5 ms pulses at 150°C)	120	V
I_D	Continuous ($T_A = 25^\circ\text{C}$, $R_{\theta JA} = 44^\circ\text{C/W}$)	1.7	A
	Pulsed (25°C , $T_{PULSE} = 300 \mu\text{s}$)	2.4	A
	Gate-to-Source Voltage	6	V
V_{GS}	Gate-to-Source Voltage	-4	V
	Recommended Gate-to-Source Voltage Operating Range*	4.5 – 5.5	V
T_J	Operating Temperature	-40 to 150	°C
T_{STG}	Storage Temperature	-40 to 150	°C

*Operating at less than 4 V_{GS} is not recommended.

Thermal Characteristics			
PARAMETER		TYP	UNIT
$R_{\theta JC}$	Thermal Resistance, Junction-to-Case	14	°C/W
$R_{\theta JB}$	Thermal Resistance, Junction-to-Board	79	
$R_{\theta JA}$	Thermal Resistance, Junction-to-Ambient (Note 1)	100	

Note 1: $R_{\theta JA}$ is determined with the device mounted on one square inch of copper pad, single layer 2 oz copper on FR4 board. See https://epc-co.com/epc/documents/product-training/AppNote_Thermal_Performance_of_eGaN_FETs.pdf for details.

Static Characteristics ($T_J = 25^\circ\text{C}$ unless otherwise stated)						
PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT	
BV_{DSS}	Drain-to-Source Voltage $V_{GS} = 0 \text{ V}$, $I_D = 125 \mu\text{A}$	100			V	
I_{DSS}	Drain-Source Leakage $V_{GS} = 0 \text{ V}$, $V_{DS} = 80 \text{ V}$		10	100	μA	
I_{GSS}	Gate-to-Source Forward Leakage $V_{DS} = 5 \text{ V}$		0.1	1	mA	
$V_{GS(TH)}$	Gate-to-Source Reverse Leakage $V_{GS} = -4 \text{ V}$		10	100	μA	
$V_{GS(TH)}$	Gate Threshold Voltage $V_{DS} = V_{GS}$, $I_D = 0.08 \text{ mA}$	0.8	1.5	2.5	V	
$R_{DS(on)}$	Drain-Source On Resistance $V_{GS} = 5 \text{ V}$, $I_D = 0.1 \text{ A}$		400	550	mΩ	
V_{SD}	Source-Drain Forward Voltage ^a $V_{GS} = 0 \text{ V}$, $I_S = 0.5 \text{ A}$		2.5		V	

^a Defined by design. Not subject to production test.

All measurements were done with substrate connected to source.



Die size: 0.9 x 0.9 mm

EPC2037 eGaN® FETs are supplied only in passivated die form with solder bumps.

Applications

- High speed DC-DC conversion
- Wireless power transfer
- Lidar/pulsed power applications
- Class-D audio

Benefits

- Ultra high efficiency
- Ultra low $R_{DS(on)}$
- Ultra low Q_g
- Ultra small footprint

Scan QR code or click link below for more information including reliability reports, device models, demo boards!



<https://lead.me/EPC2037>

eGaN® FETs for Low Cost Resonant Wireless Power Applications



Yuanzhe Zhang, Ph.D., Director of Applications Engineering, Michael de Rooij, Ph.D., Vice President of Applications Engineering

Resonant wireless power systems use loosely-coupled, highly-resonant coils that are tuned to high frequencies (6.78 MHz or 13.56 MHz). The AirFuel Alliance has developed the standard for resonant wireless power applications. They address convenience-of-use issues such as source to device distance, device orientation on the source, multiple devices on a single source, higher power capability, simplicity of use, and imperfect placement.

eGaN® FETs from EPC offer significantly lower capacitance and inductance with zero reverse recovery charge (Q_{rr}) in a smaller footprint for a given $R_{DS(on)}$ than comparable MOSFETs. This enables a number of applications that require higher switching frequency such as 6.78 MHz highly resonant wireless power transfer.

In this application note, we will present a differential class E amplifier using EPC2037 for 6.78 MHz loosely coupled highly resonant wireless power applications. A photo of the EPC2037 die is shown in figure 1, with its specifications given in table 1. It has very low gate charge and parasitics. As a result, no dedicated gate driver chips are required and it can be driven directly from logic.

Class E amplifier design basics

The schematic of a single-ended class E amplifier is shown in figure 2. It consists of a ground-referenced transistor (Q_1), an RF choke (L_{gsk}), an extra inductor (L_r) and a shunt capacitor (C_s). The load of the amplifier (Z_{load}) represents the power transmitting coil and is assumed to be inductive. A capacitor (C_c) is connected in series for coil tuning. The resistance

of the tuned coil is R_{load} . The amplifier operates at a fixed frequency with a fixed duty cycle of 50%. The transistor voltage and current waveforms under ideal operation is sketched in figure 3. Because of zero-voltage switching (ZVS) and zero-current switching (ZCS) at the optimum operating point, the class E amplifier has high efficiency (usually well above 90%).

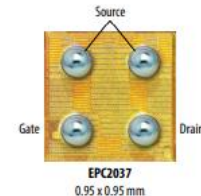


Figure 1: Mounting side of EPC2037 eGaN FET
Dimension: 0.9 x 0.9 mm

EPC Part Number	V_{DS} (V)	$R_{DS(on)}$ (mΩ)	Q_g (nC) Typ	Q_{rr} (nC) Typ	Q_{gsk} (pC) Typ	Q_{dss} (pC)	I_D (A)	Package (mm)
EPC2037	100	550	115	32	25	600	1.7	BGA 0.9 x 0.9

Table 1. EPC2037 specifications

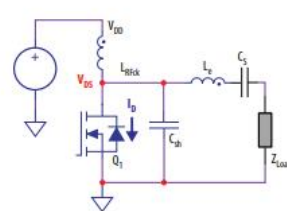


Figure 2: Schematic of a single-ended class E amplifier

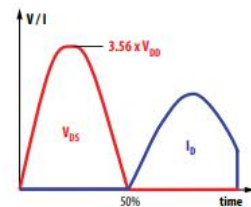


Figure 3: Ideal FET voltage and current waveforms for the class E amplifier

Results

- <https://open.library.ubc.ca/soa/cIRcle/collections/ubctheses/24/items/1.0406624>
- Paper shows a PA designed closely to our specifications at frequencies of 13.56 MHz and 27.12 MHz.
- Provides a design for how to drive the transistor and mentions need for heatsink.

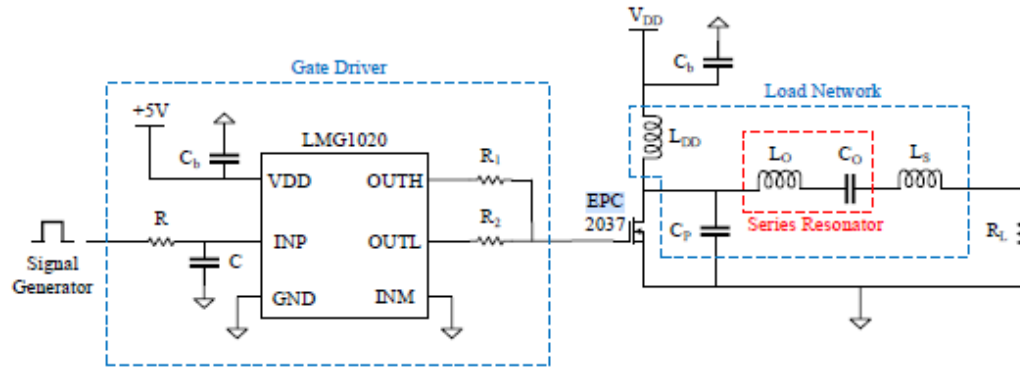


Figure 2.2: A signal generator (clock oscillator) feeds a pulse to the LMG1020 gate driver integrated with the class-E power amplifier circuit [2].

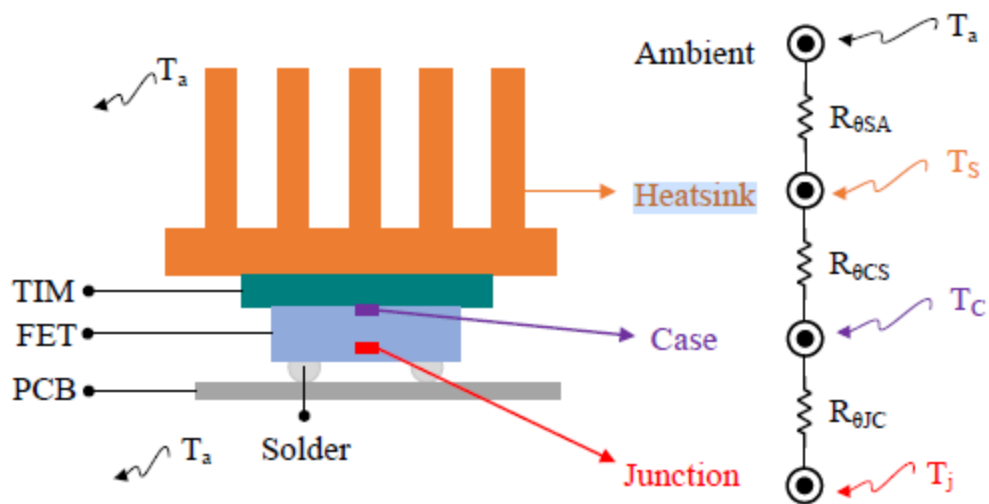


Figure 2.9: Illustration of a heatsink attachment with corresponding thermal resistance dissipation for a typical FET device.

Results

- <https://epc-co.com/epc/Portals/0/epc/documents/application-notes/AN021%20FETs%20for%20Low%20Cost%20Class%20E%20WiPo.pdf> is a document dedicated to explaining the EPC2037's application demonstrating it's use in multiple different class E PAs.

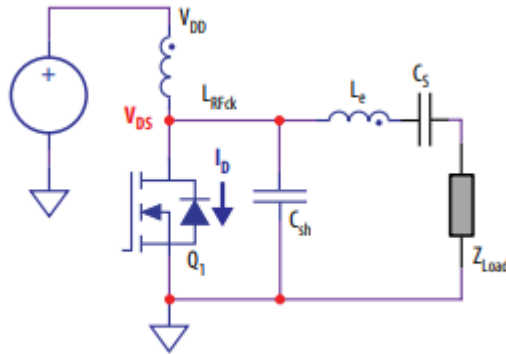


Figure 2. Schematic of a single-ended class E amplifier

Results

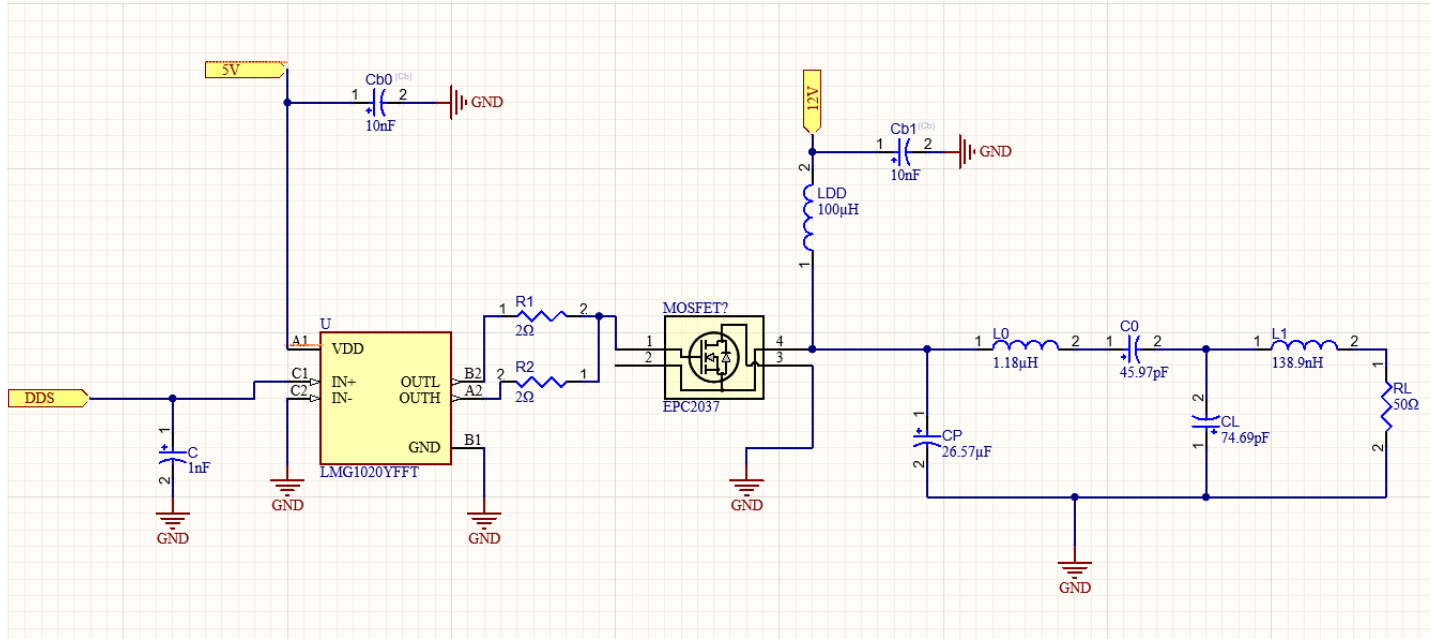


Figure 8: Altium Schematic Block implementing Driver + GaN-FET. Assumes DDS, 5V, and 12V rail are handled by MCU as it is supposed to control whether they are on or off.

Observation/Conclusion

- Previous design was lacking in power efficiency and tuned to the wrong frequency.
- The EPC2037 GaN-FET in conjunction with a driver amplifying a square wave may be preferable to 2 amplifiers in series.
 - The EPC2037 GaN-FET seems preferable over the IRF510 (due to IRF510s high output capacitance) supported by documentation and paper.
 - PCB implementation needs to take into account the inaccessible pins in the smaller device.
- Will need to collaborate in order to properly design each schematic block.

Next Steps

- Do some more testing in LTspice using the EPC2037 check power efficiency.
- Change Q value of amplifier to allow PA to accept a wider range of frequencies **efficiently** (was told it needed to be 20-25MHz). Confirm range using LTspice.
- Refine design (using real component values) and integrate it into the overall device with the help of Jeremiah and Dmytro through the use of Altium 365.

Power Amplifier Design Continued...

Luis Wong 11/04/2025

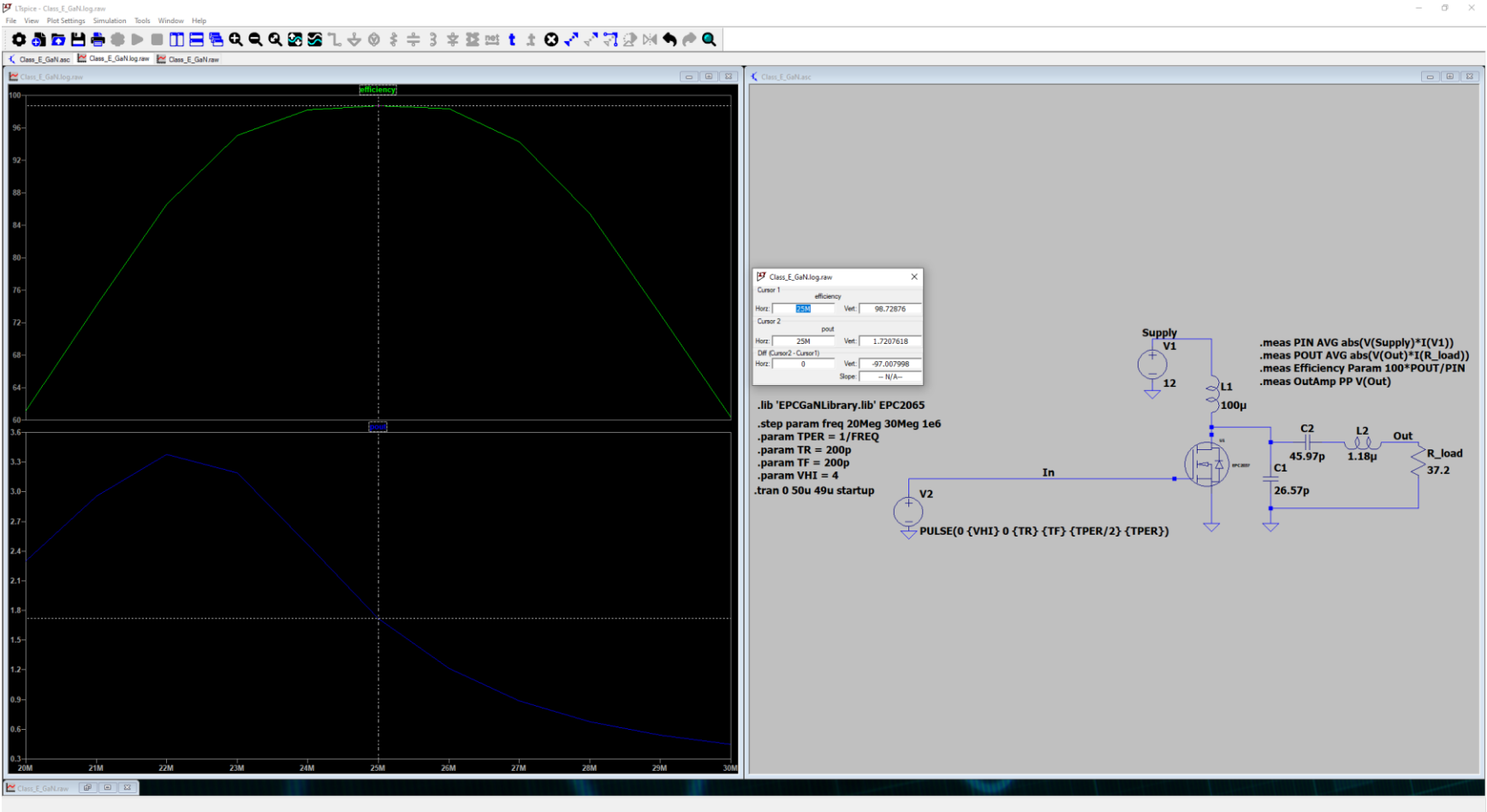
Objective

- Refine class E Design so that it may be implementable in Altium.
 - Simulate the efficiency of an ideal GaN_FET setup.
 - Achieve decent efficiency using IRF510 MOSFET.
 - Simulate 500us of the PA working.
 - User Driver in design.
 - Use capacitor and inductor values that are attainable.

Methods

- Use Altium and LTspice in order to make a working schematic and simulation.
- Use ONLINE SMITH CHART TOOL in order to make matching network.

Results



Results

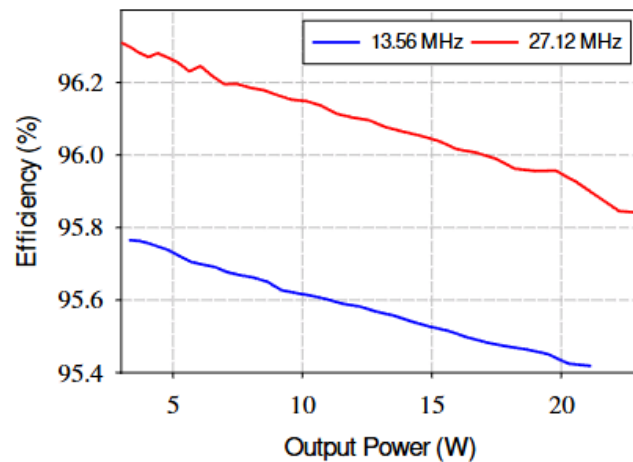


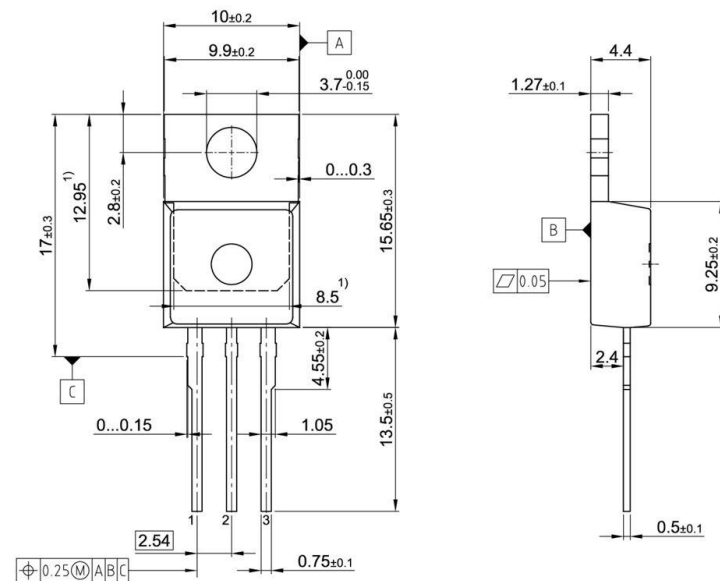
Figure 2.5: Simulated results for efficiency versus power of class-E amplifiers designed for 13.56 MHz and 27.12 MHz frequency bands.

Results



Die size: 0.9 x 0.9 mm

EPC2037 eGaN® FETs are supplied only in passivated die form with solder bumps.



1) Typical metal surface min. x=7.25 y=12.3

All dimensions are in units mm

The drawing is in compliance with ISO 128-30, Projection Method 1 [⊥]

The screenshot displays the LTSpice simulation environment. The left window shows two plots from the log file 'Class_E_rev1.log.raw'. The top plot, titled 'efficiency', shows a green curve peaking at approximately 80% efficiency around 27 MHz. The bottom plot, titled 'pout', shows a blue curve peaking at approximately 1.62 W around 24 MHz. A cursor window is overlaid on the efficiency plot, showing the following data:

Class_E_rev1.log.raw	
Cursor 1 efficiency	
Horz: 27.6M	Vert: 77.822076
Cursor 2 pout	
Horz: 27.6M	Vert: 1.6220642
Diff (Cursor2 - Cursor1)	
Horz: 0	Vert: -75.990012
Slope: -N/A-	

The right window shows the circuit diagram for a Class E power amplifier. The circuit includes a 3.7V supply (V1), an input pulse source (V2), an IRF510 MOSFET (M1), a 300pF capacitor (C1), a 500pF capacitor (C2), a 100nH inductor (L2), and a 4Ω load resistor (R_load). The circuit is simulated with the following parameters:

```

.lib 'EPGaNLibrary.lib' EPC2065
.step param freq 20Meg 30Meg 1e6
.param TPER = 1/FREQ
.param TR = 200p
.param TF = 200p
.param VHI = 10
.tran 0 50u 49u startup

```

The simulation results are displayed in the bottom status bar: **x = 26.26M v = 65.91**.

$x = 26.26\text{M}$ $y = 65.91$

Results

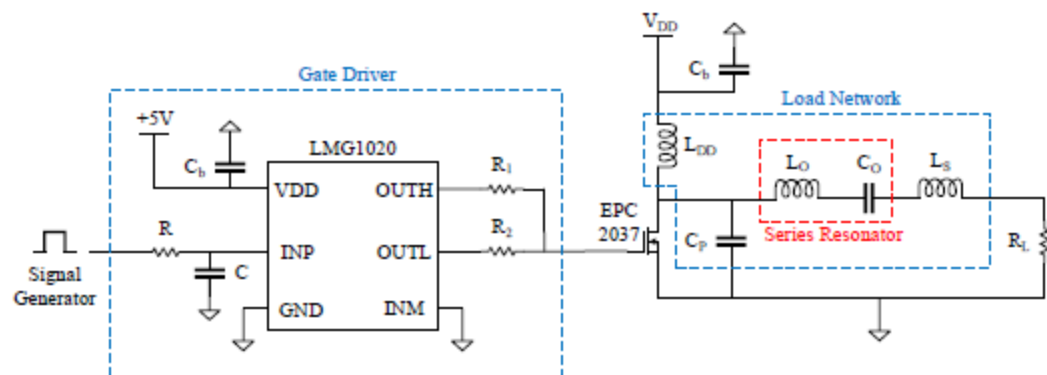
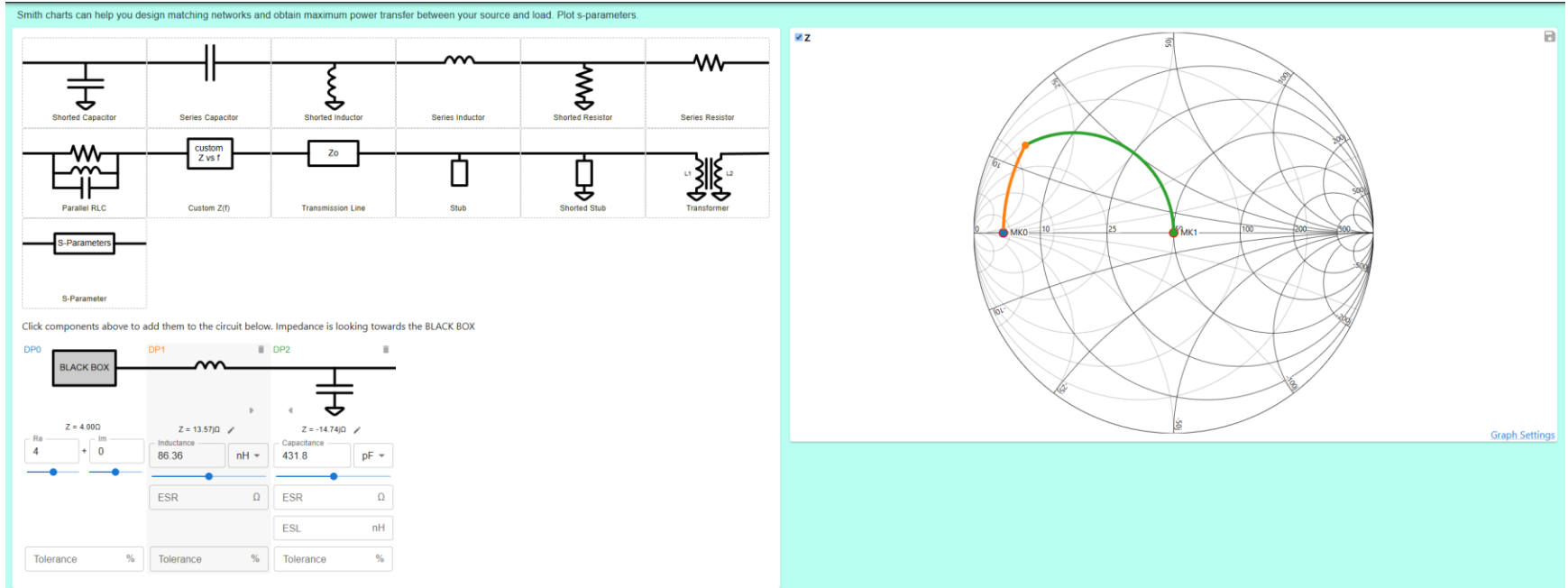


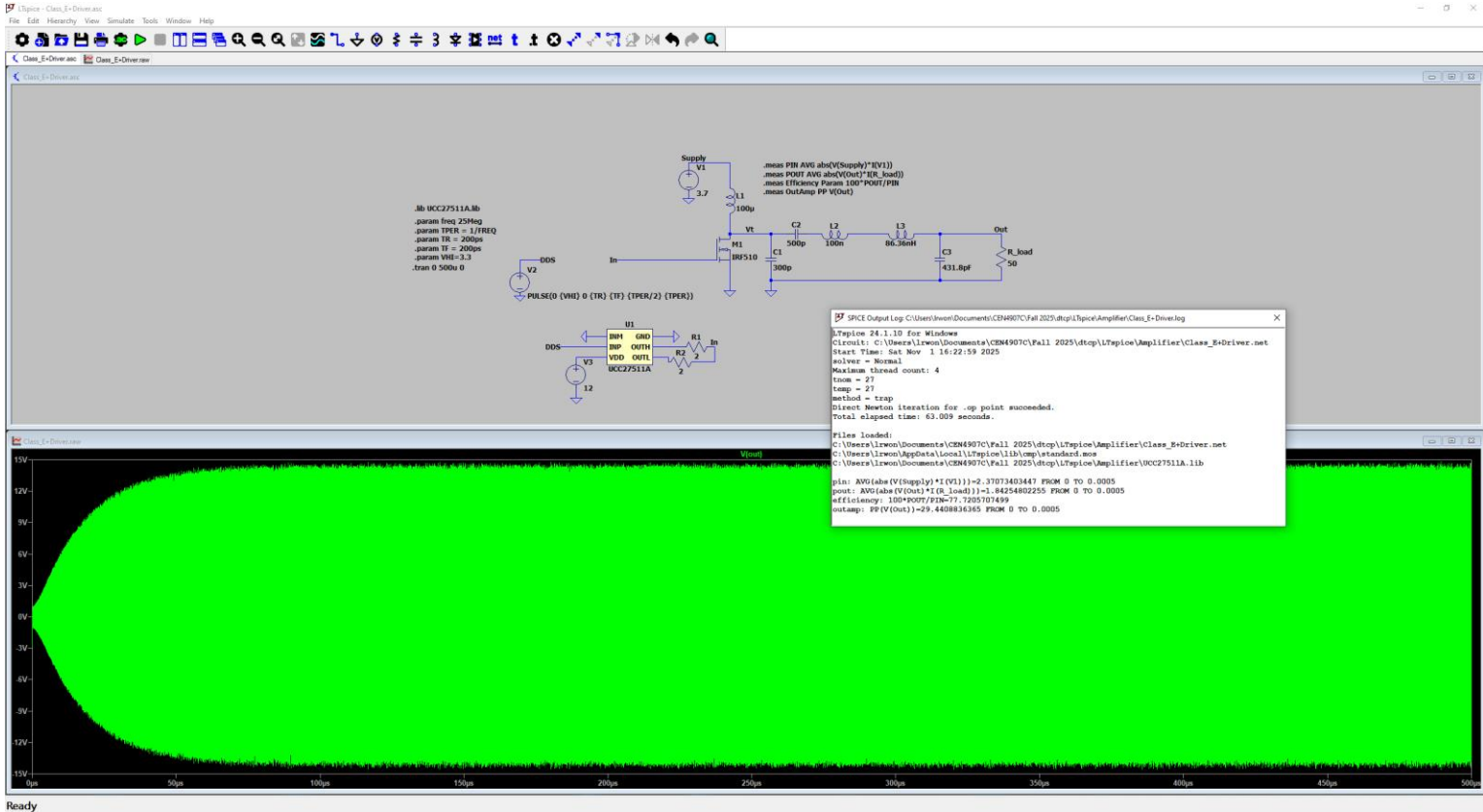
Figure 2.2: A signal generator (clock oscillator) feeds a pulse to the LMG1020 gate driver integrated with the class-E power amplifier circuit [2].

Results

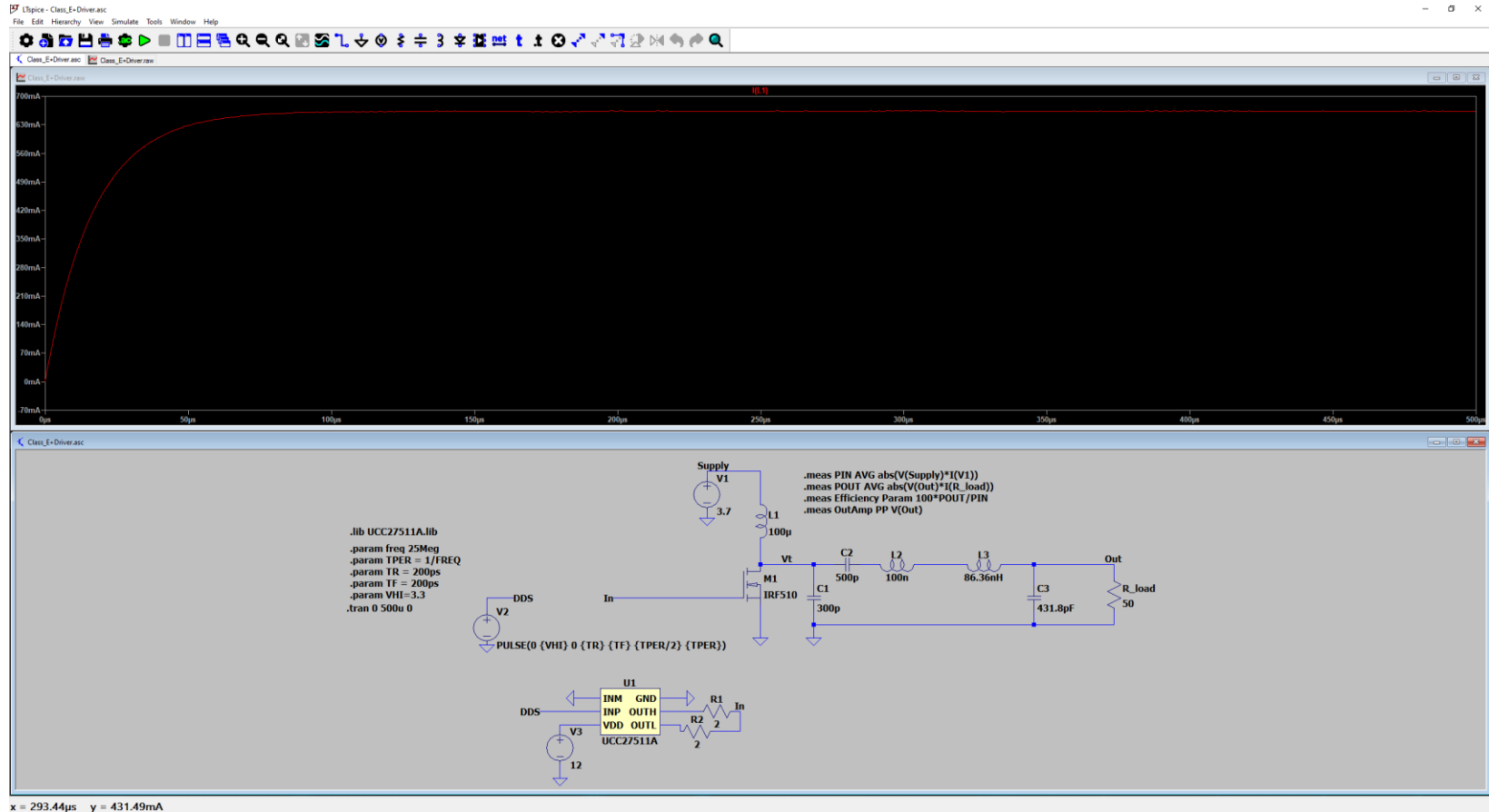


Will use calculator in order to tune matching network (an approx. 4 ohm to 50 ohm impedance).

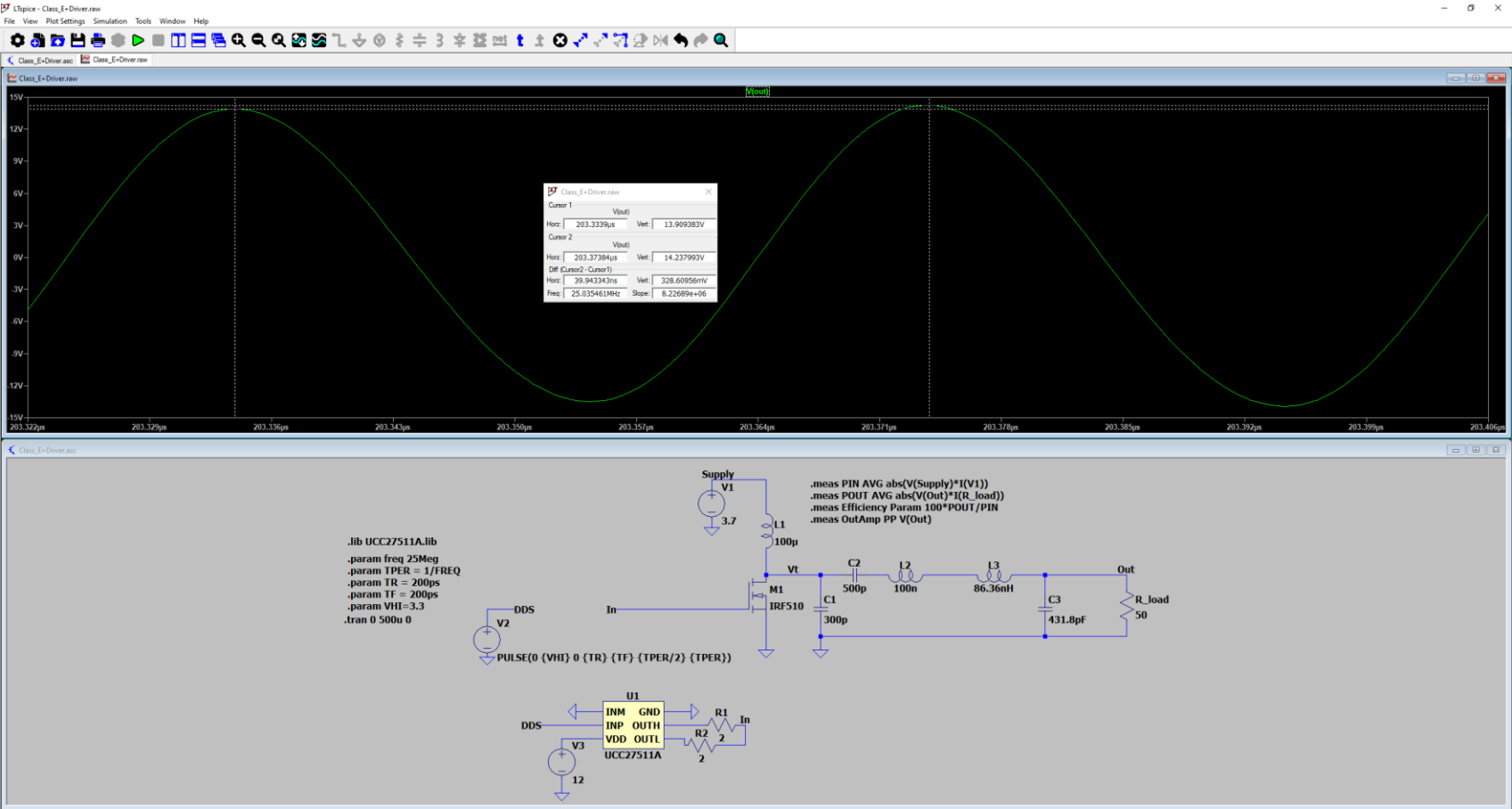
Results (DDS: 3.3V Square Wave)



Results (DDS: 3.3V Square Wave)



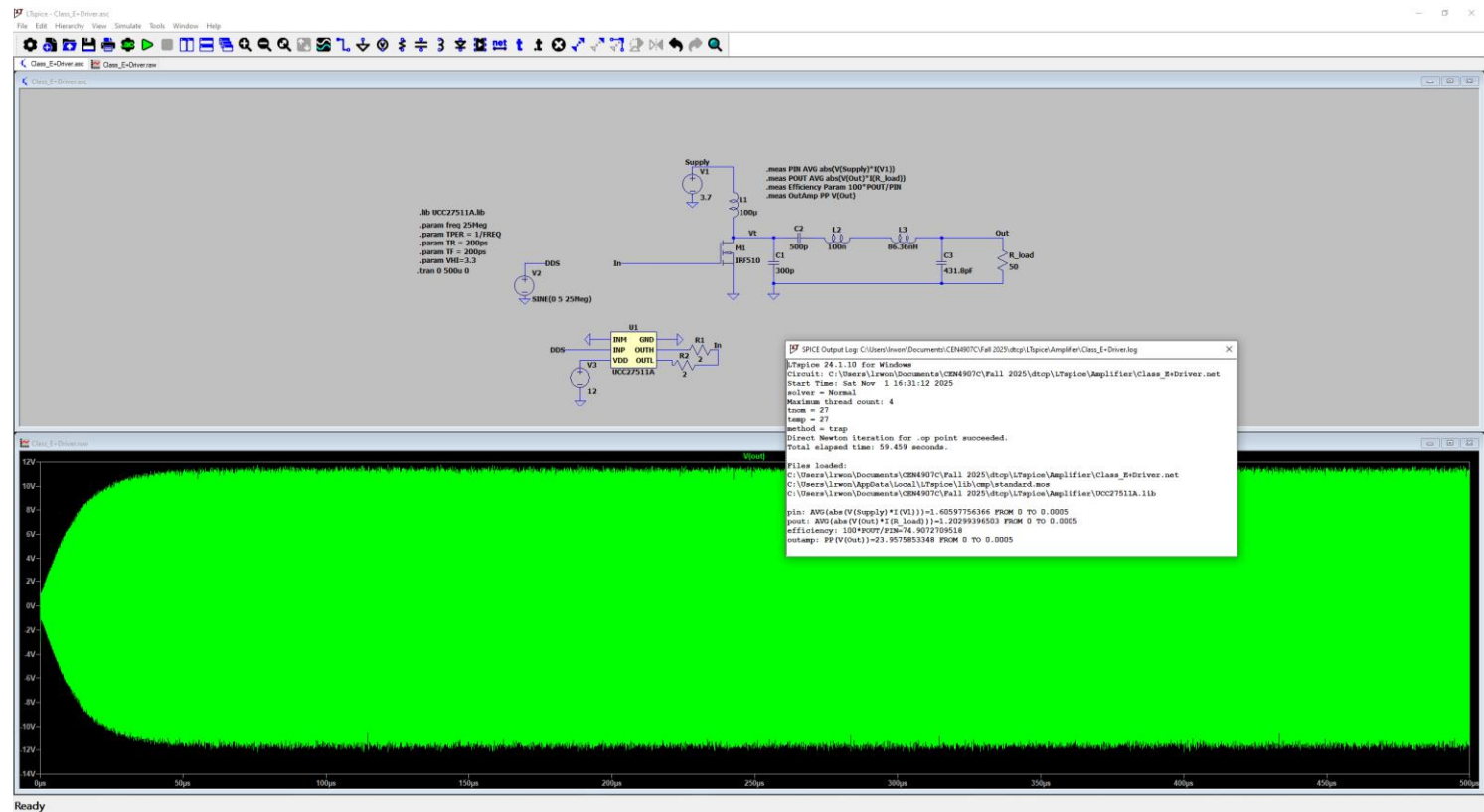
Results (DDS: 3.3V Square Wave)



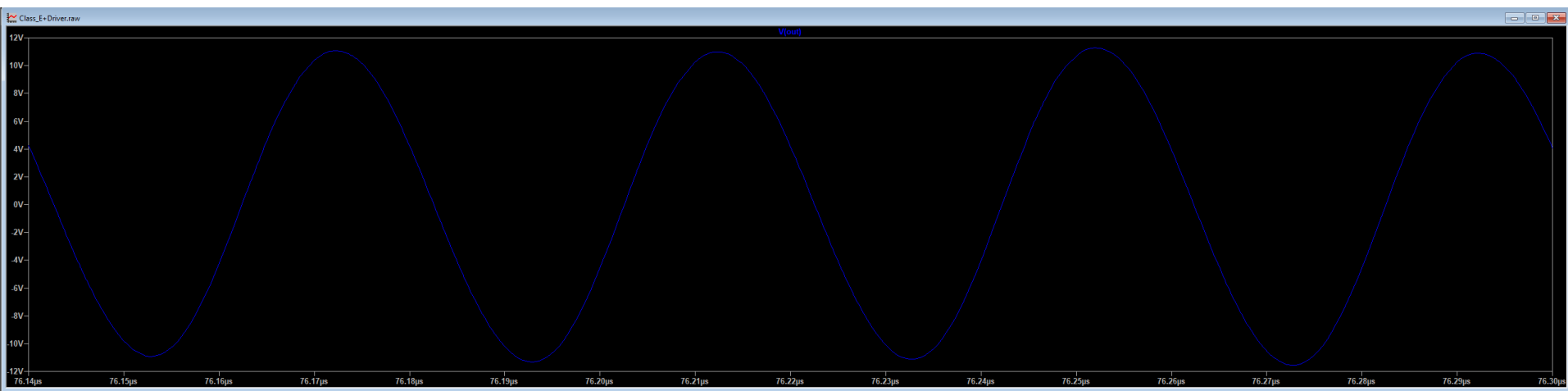
x = 203.33199us y = -4.04V

Results (DDS: 5V Sine Wave)

3.3 V makes power out go down to 0.7W

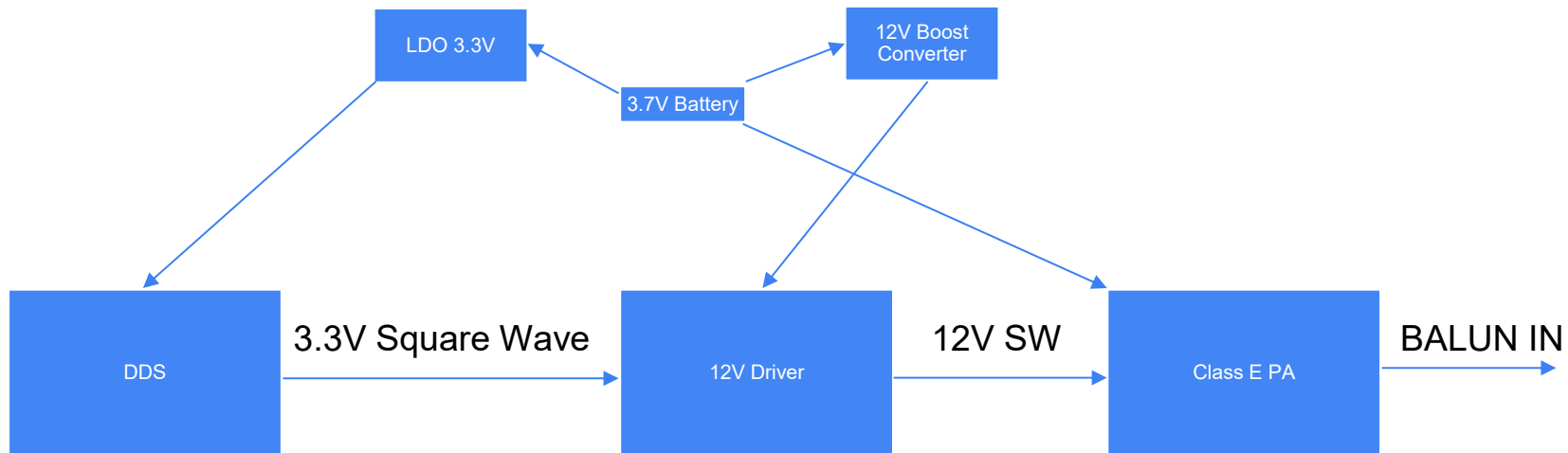


Results



This results in distorted sine wave as output a period of about 50ns instead of the 40ns we expect (Each tick is 10ns).

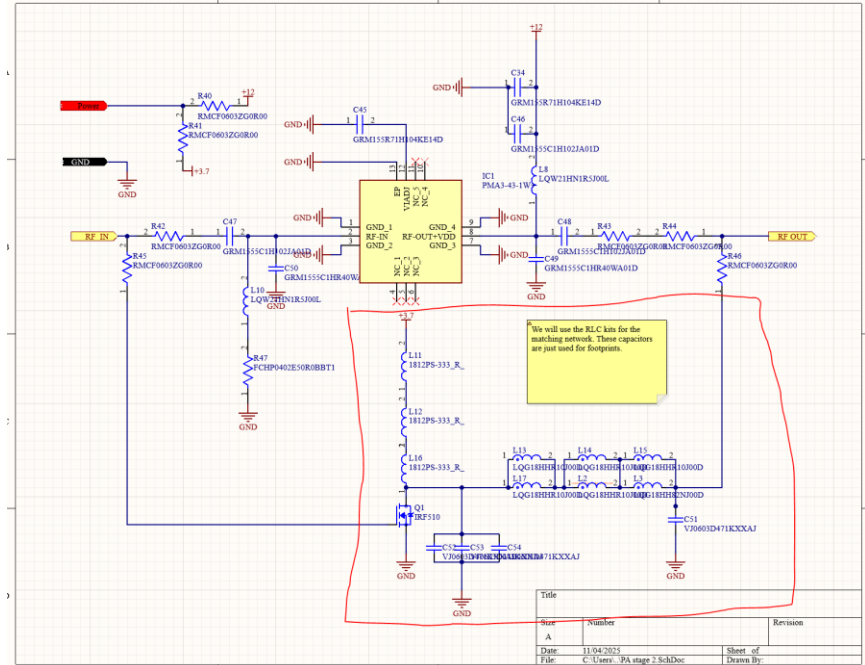
Results



Bill of Material - TQP369184-PCB

Reference Des.	Value	Description	Manufacturer	Part Number
Q1	0.01 uF	High Linearity LNA Gain Block	Gamex	TQP369184
C1	0.01 uF	Cap. Chip, 0603, 10V, X7R, 10%	Various	
C2, C3	68 pF	Cap. Chip, 0603, 50V, APO, 5%	Various	
L1	39 nH	Inductor, 0603, 5%, CS Series	Coilcraft	0603CS-39NHL
R1	24 Ohm	Res. Chip, 0603, 1/10W, 5%	Various	

The circuit diagram shows a multi-stage PCB design. It includes an RF input stage with components R36, C14, Q1, C1, C2, C3, L1, and R1. The signal path continues through several matching networks (R38, R39, R40) and various capacitors (C15, C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, C26, C27, C28, C29, C30, C31, C32, C33, C34, C35, C36, C37, C38, C39, C40, C41, C42, C43, C44, C45, C46, C47, C48, C49, C50, C51, C52, C53, C54, C55, C56, C57, C58, C59, C60, C61, C62, C63, C64, C65, C66, C67, C68, C69, C70, C71, C72, C73, C74, C75, C76, C77, C78, C79, C80, C81, C82, C83, C84, C85, C86, C87, C88, C89, C90, C91, C92, C93, C94, C95, C96, C97, C98, C99, C100). The output stage features a power amplifier (Q2) and a matching network (R41, R42, R43, R44, R45, R46, R47, R48, R49, R50, R51, R52, R53, R54, R55, R56, R57, R58, R59, R60, R61, R62, R63, R64, R65, R66, R67, R68, R69, R70, R71, R72, R73, R74, R75, R76, R77, R78, R79, R80, R81, R82, R83, R84, R85, R86, R87, R88, R89, R90, R91, R92, R93, R94, R95, R96, R97, R98, R99, R100). The PCB is populated with various passive components, including resistors (R1-R100), capacitors (C1-C100), and inductors (L1-L100).



Observation/Conclusion

- As we are approaching the end of the semester the IRF510 while not as efficient is easier to prototype with.
 - Manages the >50% efficiency Jeremiah was looking for.
- Altium Schematic was implemented.

Next Steps

- Correct any mistakes in Altium design and adjust based on feedback.
- Adjust matching network design to work with real values.
- Order PCB in order to begin firmware development
- Assemble PCB and test functionality.

Inductors and Simulating New GaN-FET

Luis Wong 11/04/2025

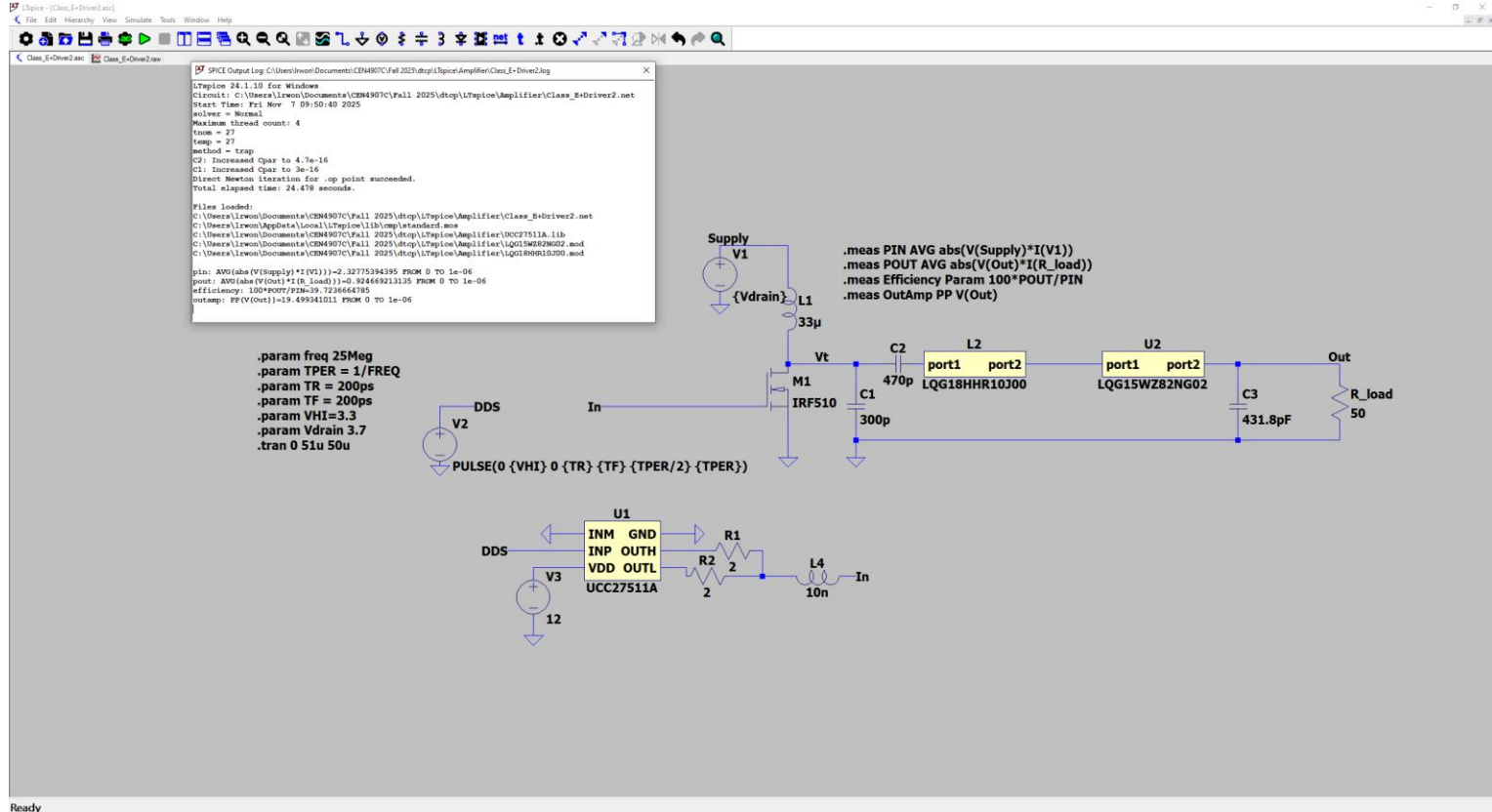
Objective

- Simulate Class-E PA with all real parts.
- If efficiency loss is present find out why.
- Fix it if possible.

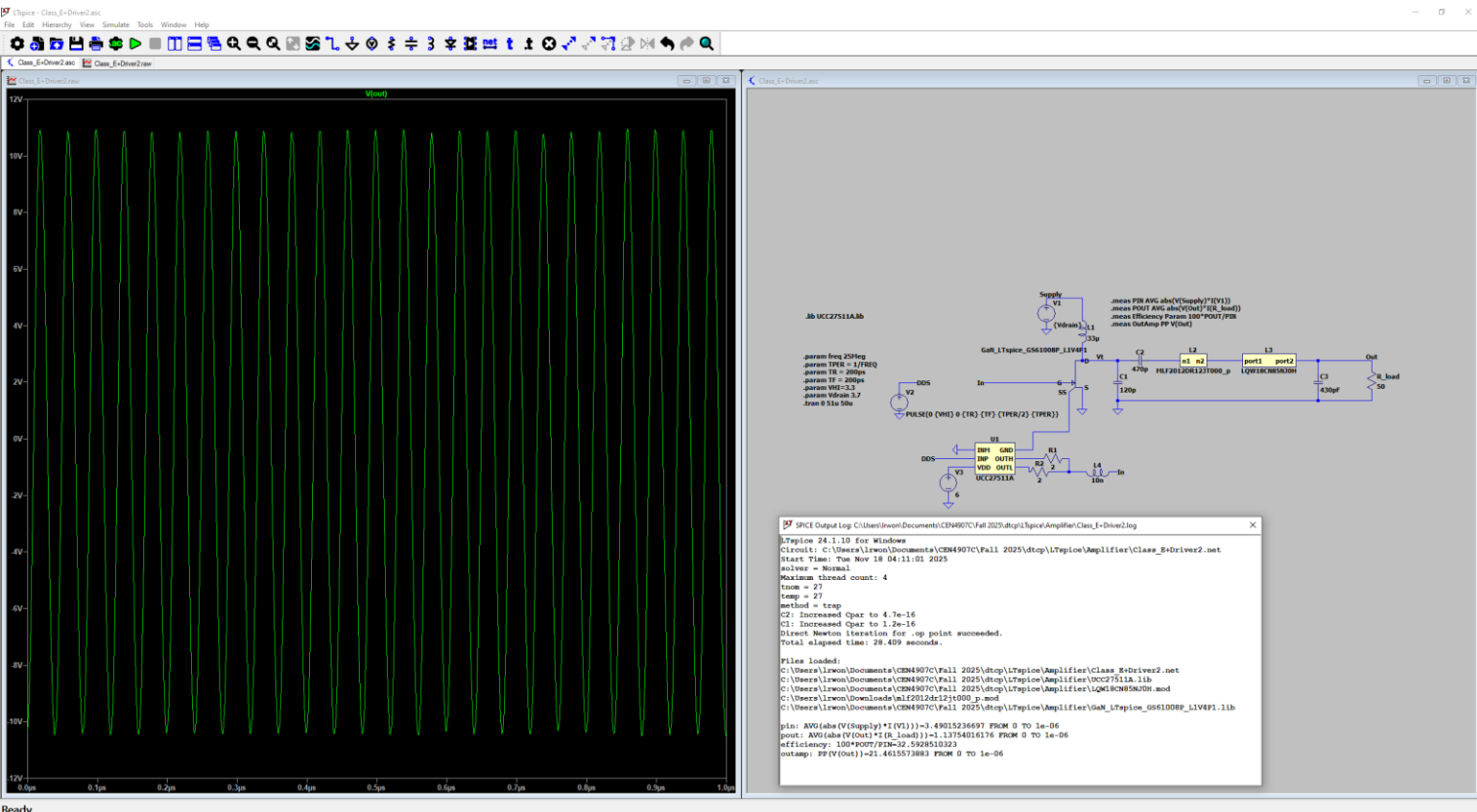
Methods

- Use LTspice and spec sheets.

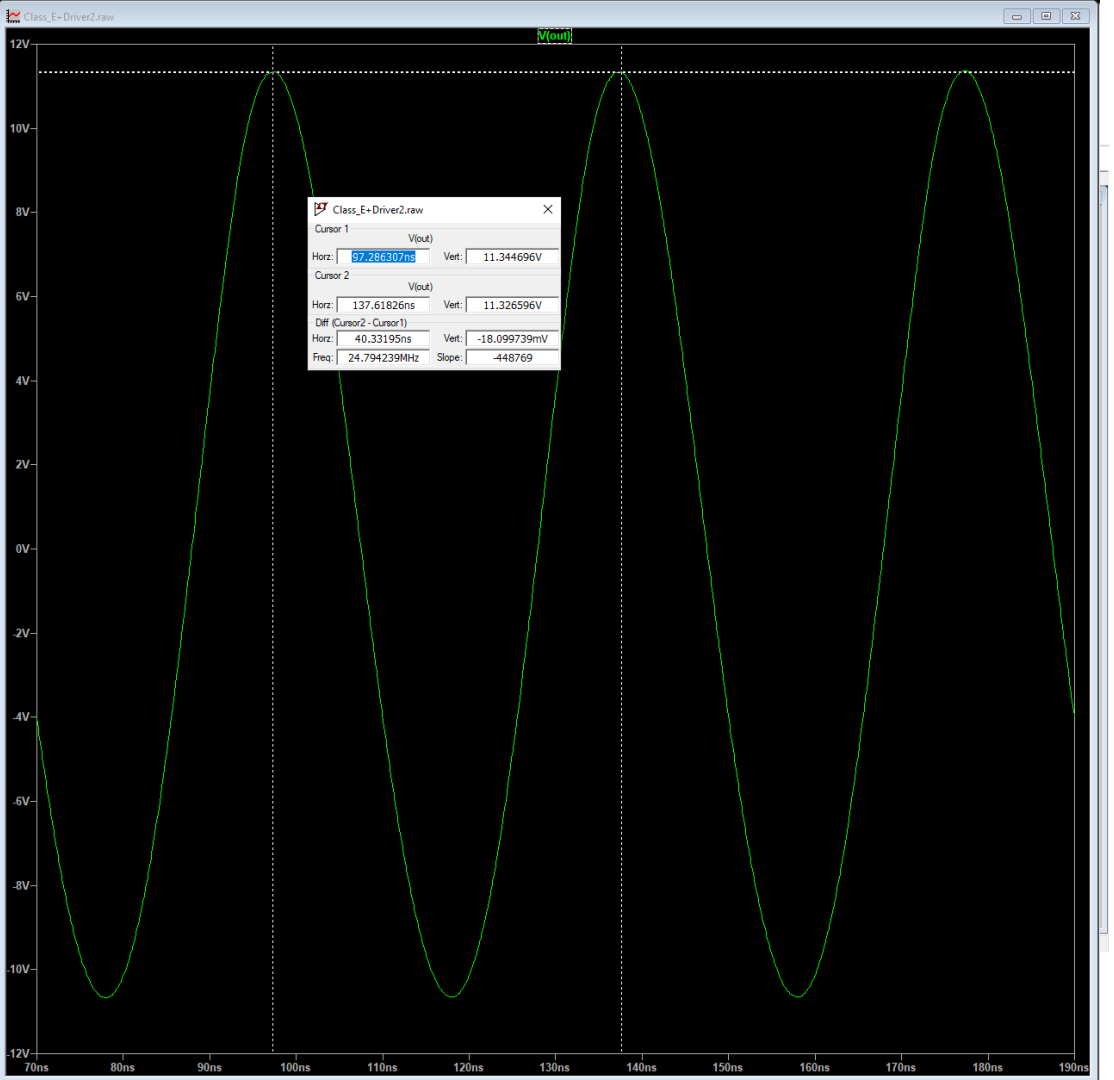
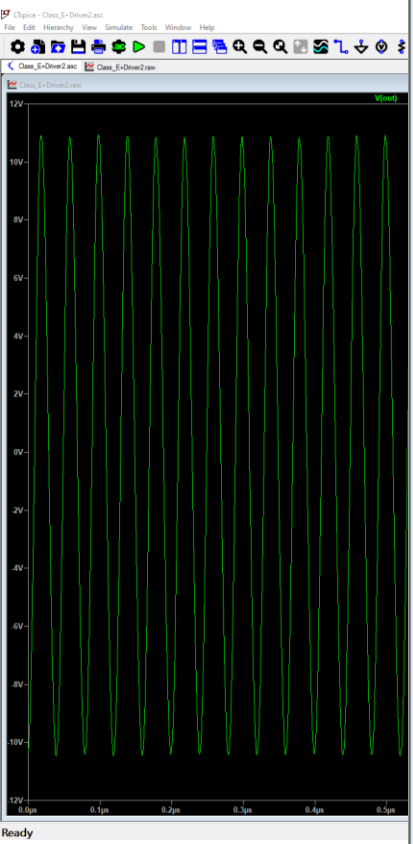
Results



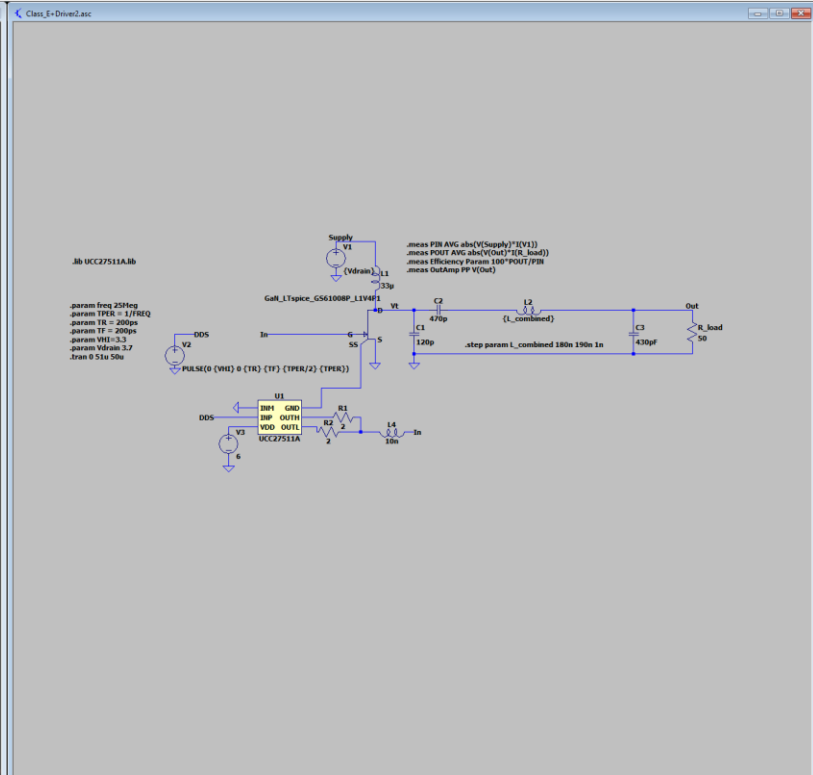
Results



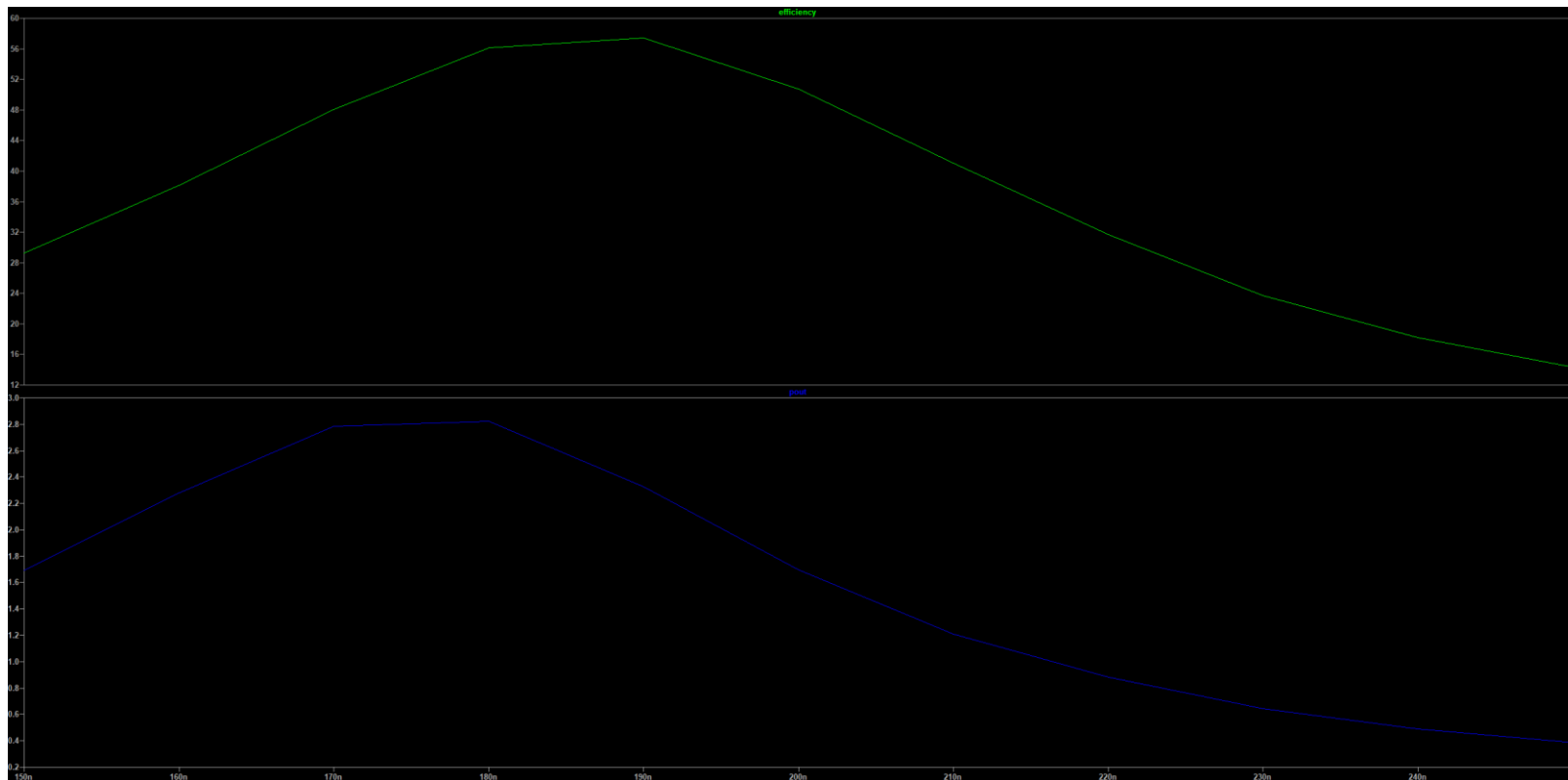
Results



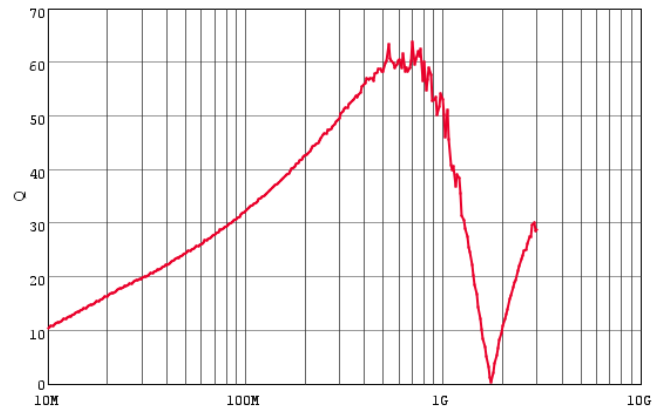
Results



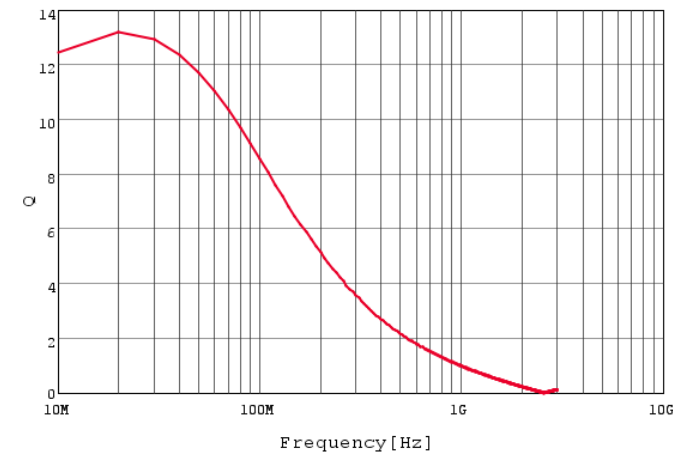
Results



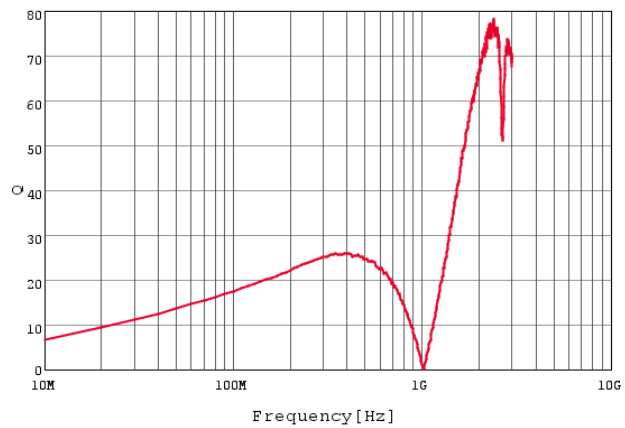
Results



LQW18ANR11G8Z Q by Imp. Analyzer

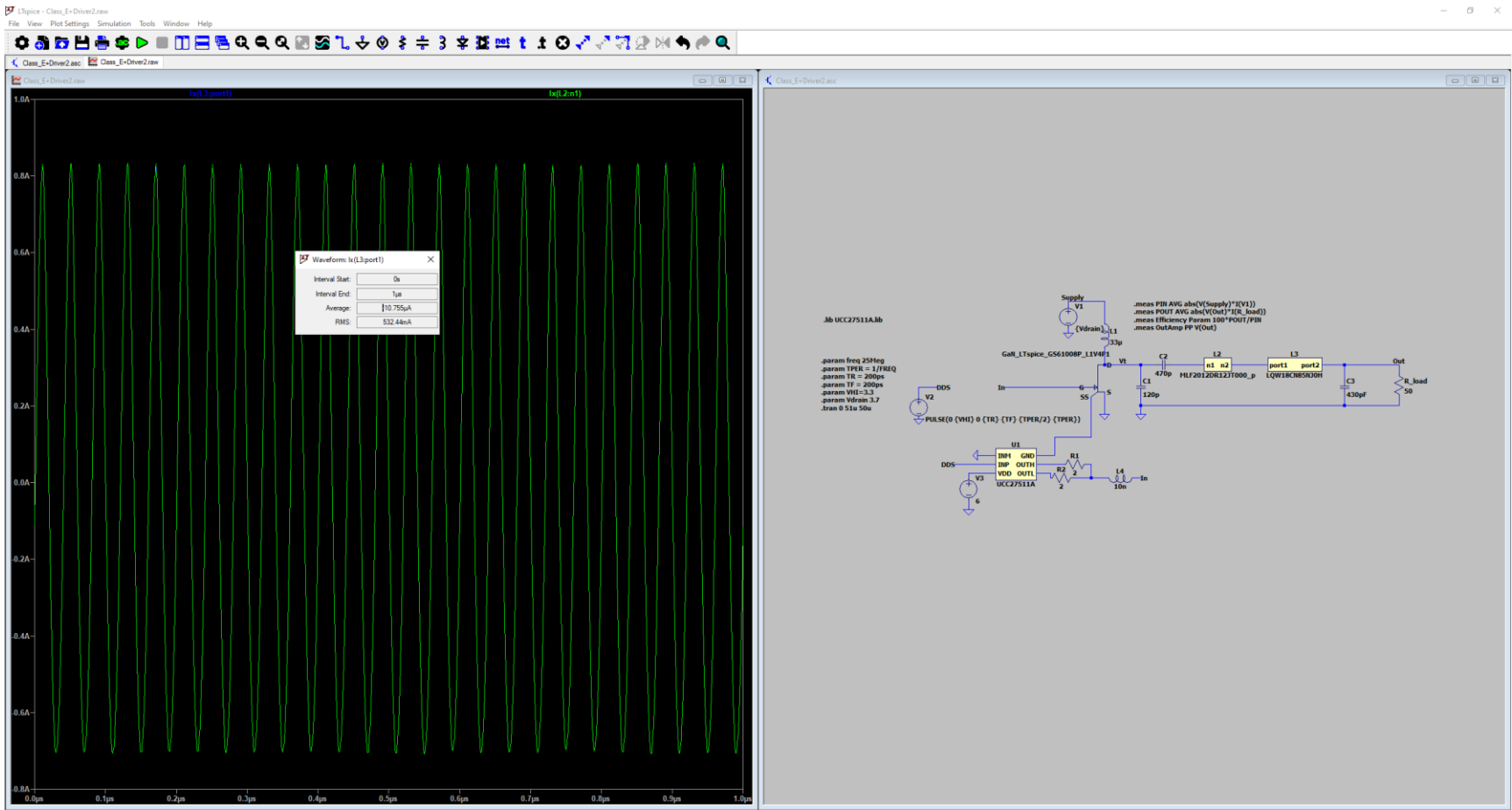


LQW18CN85NJ0H Q



LQG18HHR10J00 Q by Imp. Analyzer

Results



Left-Click to redraw $i(L3.port1)[A]$. Right-Click to edit. Control-Left-Click to integrate. Alt-Left-Click to reverse cross probe L3.

Results

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Age (CT) Reel & Reel (TR)	Last Time Buy	Drum Core, Wirewound Multilayer Wirewound	Ferrite Iron	±5% ±10% ±20%	650 mA 800 mA 1.018 A 1.12 A 1.131 A 1.89 A 2.85 A	700mA -	Shielded Unshielded	24mOhm Max 48mOhm Max 90mOhm Max 150mOhm Max 180mOhm Max 200mOhm Max 512 SmOhm Max	5 @ 25 MHz 8 @ 25MHz 10 @ 25 MHz 15 @ 25MHz 30 @ 25MHz 35 @ 25 MHz 40 @ 25MHz	200MHz 300MHz 375MHz 383MHz 450MHz 550MHz 780MHz 800MHz	- AEC-Q200	-55°C ~ 125°C -40°C ~ 100°C -40°C ~ 105°C -40°C ~ 125°C -40°C ~ 85°C	25 MHz 25.2 MHz	0603 (160) 1008 (252) 1210 (322) 1812 (453)

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Mfr Part #	Quantity Available	Price	Tariff Status	Series	Package	Product Status	Type	Material - Core	Inductance	Tolerance	Current Rating (Amps)	Current - Saturation (I _{sat})	Shielding	DC Resistance (DCR)	Q @ Freq	Frequency - Self Resonant	Rated Power	Operating Temperature	Inductance Frequency - Test
 S1210R-101K FIXED IND 100NH 1.131A 150 MOHM View all quantities Daletron	1,819 In Stock	1 : \$3.59000 Cut Tape (CT) 500 : \$2.09462 Tape & Reel (TR)	-	S1210R	Tape & Reel (TR) Ⓢ Cut Tape (CT) Ⓢ Dig-Reel® Ⓢ	Active	-	Iron	100 nH	±10%	1.131 A	-	Shielded	150mOhm Max	40 @ 25MHz	275MHz	-	-55°C ~ 125°C	25 MHz
 S1008R-101K FIXED IND 100NH 1.12A 90MOHM SMD View all quantities Daletron	1,421 In Stock	1 : \$5.91000 Cut Tape (CT) 500 : \$1.76718 Tape & Reel (TR)	-	S1008R	Tape & Reel (TR) Ⓢ Cut Tape (CT) Ⓢ Dig-Reel® Ⓢ	Active	-	Ferrite	100 nH	±10%	1.12 A	-	Shielded	90mOhm Max	40 @ 25MHz	383MHz	-	-50°C ~ 125°C	25 MHz
 CM45232-810ML FIXED IND 100NH 800MA 180MOHM SMD View all quantities Bourns Inc	1,102 In Stock	1 : \$8.27000 Cut Tape (CT) 500 : \$0.15748 Tape & Reel (TR)	Tariff may apply if shipping to the United States	CM45	Tape & Reel (TR) Ⓢ Cut Tape (CT) Ⓢ Dig-Reel® Ⓢ	Active	Drum Core, Wirewound	Ferrite	100 nH	±20%	800 mA	-	Unshielded	180mOhm Max	35 @ 25 MHz	300MHz	-	-40°C ~ 125°C	25.2 MHz
 PM1812-B10K-BC FIXED IND 100NH 800MA View all quantities	500 In Stock	1 : \$8.20000 Cut Tape (CT)	Tariff may apply if shipping to the United States	PM1812	Tape & Reel (TR) Ⓢ Cut Tape (CT) Ⓢ	Active	-	Ferrite	100 nH	±10%	800 mA	-	Unshielded	180mOhm Max	35 @ 9KHz	300MHz	-	-40°C ~ 100°C	25.2 MHz

[Feedback](#)

Results

Products Magnetic Products, Power Inductors - SMD Non-shielded

Model	Description
SDE0403AT	SMD Power Inductors
SDE6603	SMD Power Inductors
SDR0302	SMD Power Inductors
sdr0403	SMD Power Inductors
SDR0503	SMD Power Inductors
SDR0603	SMD Power Inductors
SDR0604	SMD Power Inductors
SDR0703	SMD Power Inductors
SDR0805	SMD Power Inductors
SDR0906	SMD Power Inductors
SDR1005	SMD Power Inductors
SDR1006	SMD Power Inductors
SDR1030	SMD Power Inductors
SDR1045	SMD Power Inductors
sdr1105	SMD Power Inductors
sdr1305	SMD Power Inductors
SDR1307	SMD Power Inductors
SDR1806	SMD Power Inductors
SDR2207	SMD Power Inductors
SDR6603	SMD Power Inductors
SDR7030	SMD Power Inductors
SDR7045	SMD Power Inductors

The screenshot displays the Vishay website's design tools section. The top navigation bar includes links for PRODUCTS, APPLICATIONS, RESOURCES, CUSTOM CAPABILITIES, and COMPANY. Below the navigation bar, the 'DESIGN TOOLS' section is visible, featuring a 'Product Category' dropdown menu. The 'Design Tool Type' section shows a list of components: S-Parameters, Diodes and Rectifiers, Discrete Thyristors, ICA - Power and Linear, MOSFETs, Optoelectronics, and Power Modules. The 'Passive Components' section includes Capacitors, Magnetics, Other Components, Resistors, and Sensors. A 'Reset All' button is present. Below the tool selection, a table lists the design files for SMD Inductors, showing columns for Design Tool Type, Description, and Product Category. The table lists 11 files, including S-Parameters, RFLYV S-Parameter Files, and RFLYV S-Parameter Files.

Design Tool Type	Description	Product Category
S-Parameters	RFLYV S-Parameter Files (*.zip)	RF Inductors - Surface Mount

Bourns Website Showing All Design File Listings

Vishay Website Showing All Design File Listings For SMD Inductors

Observation/Conclusion

- The ~30% efficiency we see is probably that low due to low Q of parts at 25MHz.
- Finding inductors that are SMDs, fit our required inductance, are listed with their Q values at 25MHz, and have a readily available spice file is difficult.
 - It seems similar implementations of Class-E Amplifiers make use of hand wound inductors.
- If we would like to test these higher Q inductors, we will need to probably buy them to test them on the physical board.

Assembling PCB

Luis Wong 01/21/2026

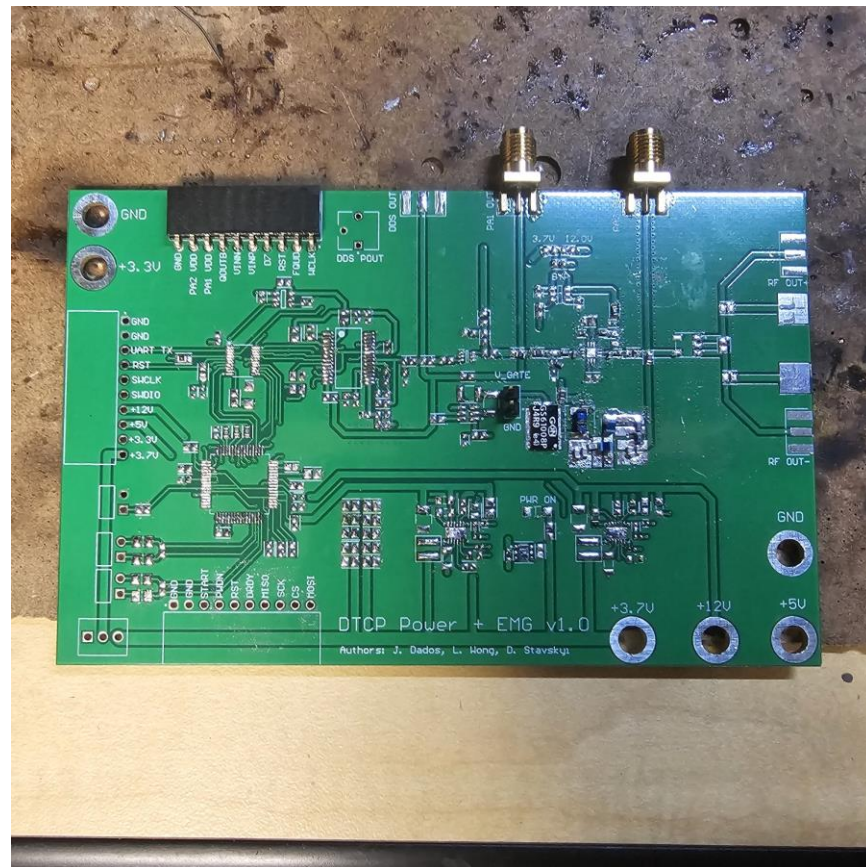
Objective

- Build the Class-E PA
- Test to see if we get the desired output at 30dBm.

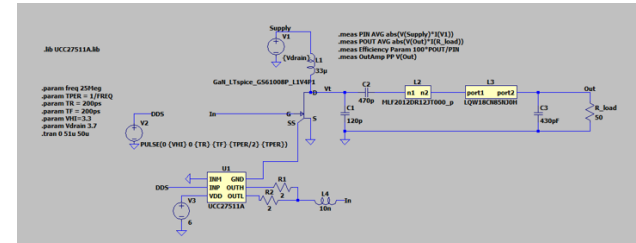
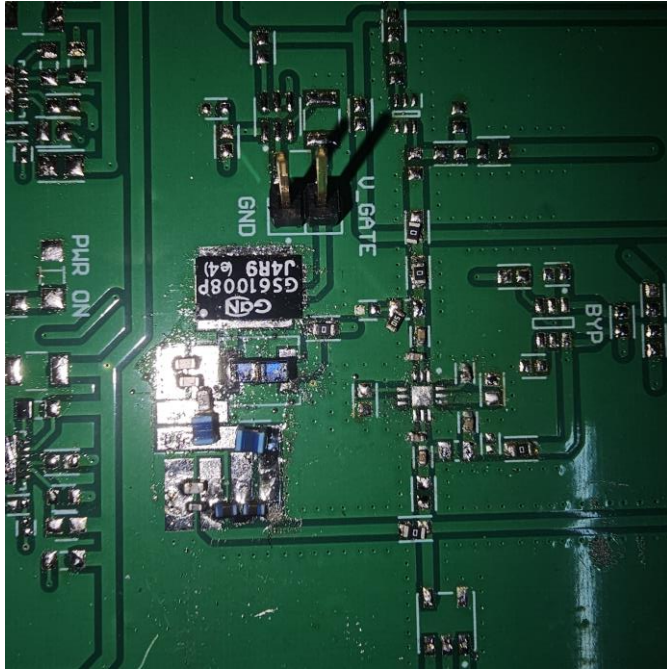
Methods

- Use spectrum analyzer, PSU, and assembled PCB to test whether the output is about 30dBm at 25MHz.

Results



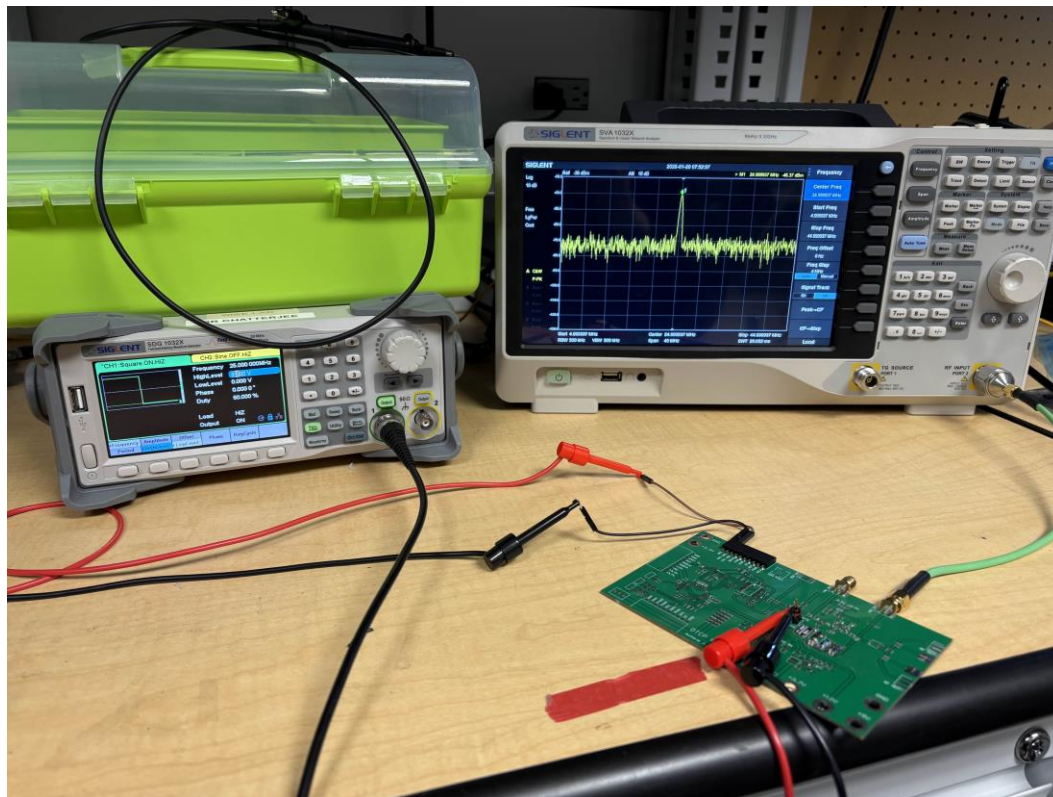
Results



We made use of parts on hand to approximate the values for the capacitors and inductors used in simulation.

- Drain inductor used was 1.5uH.
- Approximated 120pF to 112pF using two 56pF capacitors in parallel.
- Approximated a 120nH and 85nH in series with two 68nH and two 39nH inductors in series.
- Used a single 470pF capacitor and a 330pF and 100pF capacitor in parallel to make a 430pF equivalent capacitor.

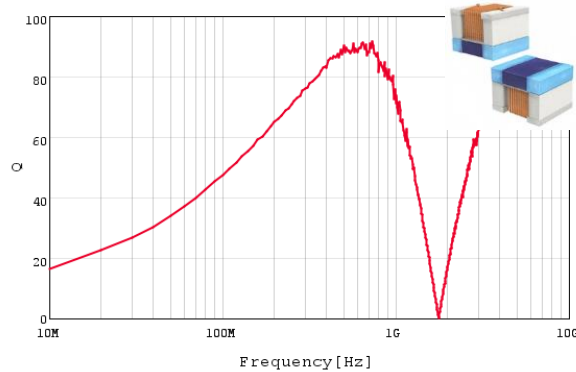
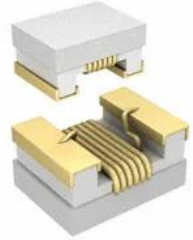
Results



Results (Checking Inductors Used)

Electrical Properties:

Properties	Test conditions		Value	Unit	Tol.
Inductance	250 MHz	L	39	nH	±5%
Q-Factor	250 MHz	Q_{min}	36		min.
DC Resistance	@ 20 °C		0.21	Ω	max.
Rated Current	$\Delta T = 15$ K	I_R	600	mA	max.
Self Resonant Frequency		f_{res}	2200	MHz	min.

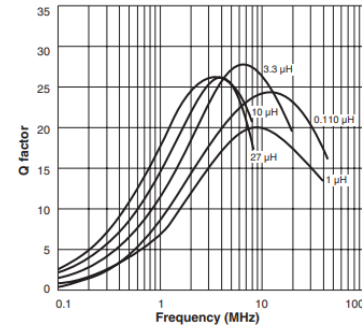


LQW2BAN68NG00# Q

Datasheet for the 68NH 1.25A
200mOHM SMD

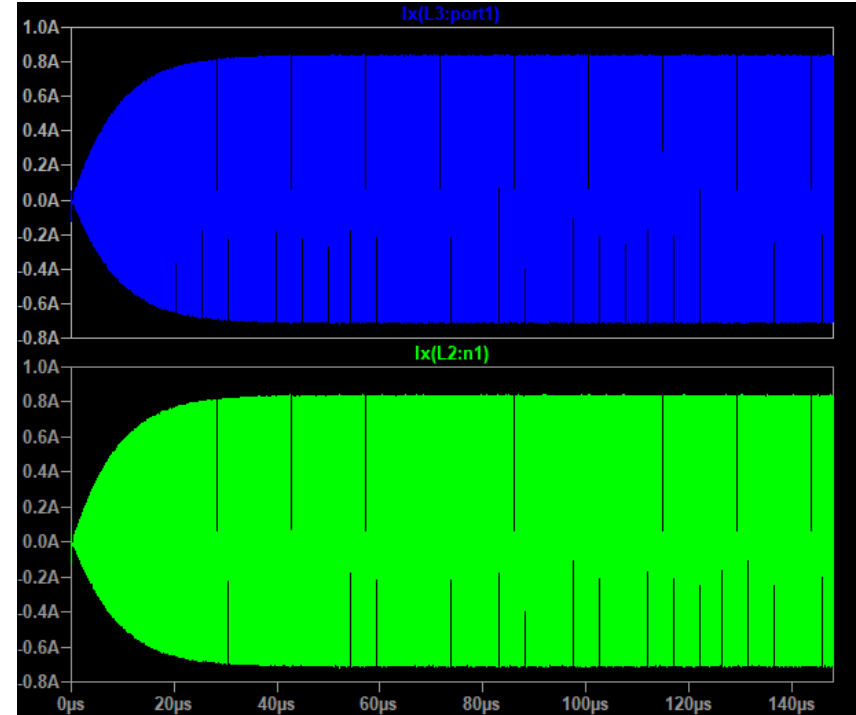
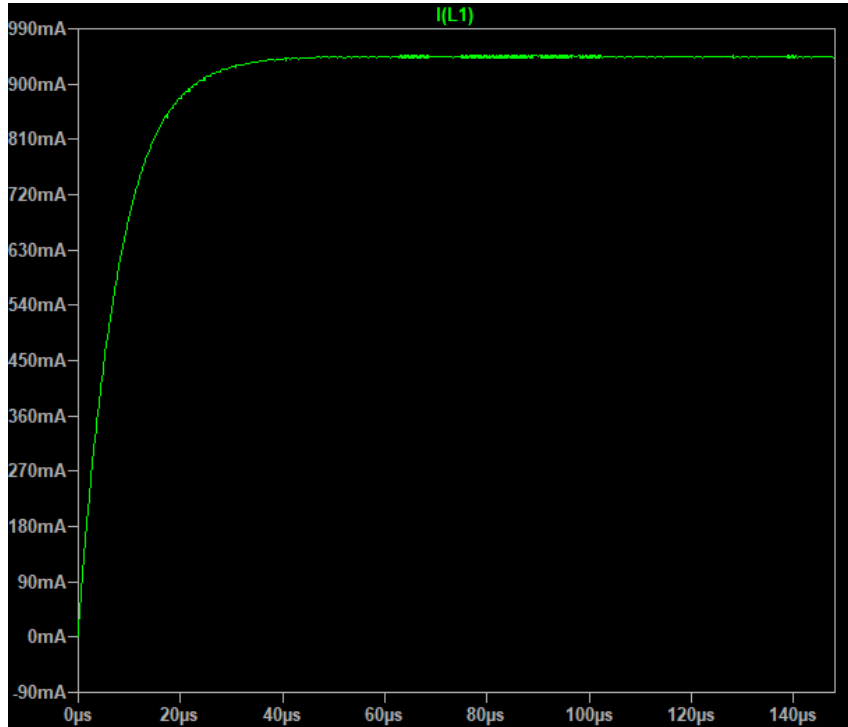
Datasheet for the 744761139C: 39NH 600MA 210MOHM SMD

Typical Q vs Frequency



Datasheet for the 0805LS-152XJRC: 1.5UH 490MA 210MOHM SMD

Results (Theoretical Currents)



Results (Checking Capacitors Used)

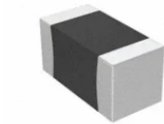


Image shown is a representation only. Exact specifications should be obtained from the product data sheet.



CC0402KRX7R9BB331

DigiKey Part Number

311-1027-2-ND - Tape & Reel (TR)
311-1027-1-ND - Cut Tape (CT)
311-1027-6-ND - Digi-Reel®

Manufacturer [YAGEO](#)

Manufacturer Product Number CC0402KRX7R9BB331

Description CAP CER 330PF 50V X7R 0402

Manufacturer Standard Lead Time 16 Weeks

Customer Reference

Detailed Description

330 pF ±10% 50V Ceramic Capacitor X7R 0402 (1005 Metric)

Datasheet

[Datasheet](#)

EDA/CAD Models

[CC0402KRX7R9BB331 Models](#)

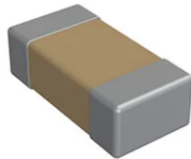


Image shown is a representation only. Exact specifications should be obtained from the product data sheet.

C0603C101K4GACTU

DigiKey Part Number

399-17390-2-ND - Tape & Reel (TR)
399-17390-1-ND - Cut Tape (CT)
399-17390-6-ND - Digi-Reel®

Manufacturer [KEMET](#)

Manufacturer Product Number C0603C101K4GACTU

Description CAP CER 100PF 16V C0G/NP0 0603

Manufacturer Standard Lead Time 14 Weeks

Customer Reference

Detailed Description

100 pF ±10% 16V Ceramic Capacitor C0G, NP0 0603 (1608 Metric)

Datasheet

[Datasheet](#)

EDA/CAD Models

[C0603C101K4GACTU Models](#)

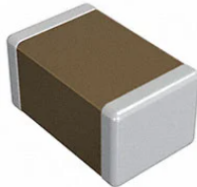


Image shown is a representation only. Exact specifications should be obtained from the product data sheet.

GRM1885C1H560JA01D

DigiKey Part Number

490-1421-2-ND - Tape & Reel (TR)
490-1421-1-ND - Cut Tape (CT)
490-1421-6-ND - Digi-Reel®

Manufacturer [Murata Electronics](#)

Manufacturer Product Number GRM1885C1H560JA01D

Description CAP CER 56PF 50V C0G/NP0 0603

Customer Reference

Detailed Description

56 pF ±5% 50V Ceramic Capacitor C0G, NP0 0603 (1608 Metric)

Datasheet

[Datasheet](#)

EDA/CAD Models

[GRM1885C1H560JA01D Models](#)



Image shown is a representation only. Exact specifications should be obtained from the product data sheet.

VJ0603D471KXXAJ

DigiKey Part Number

720-1335-2-ND - Tape & Reel (TR)
720-1335-1-ND - Cut Tape (CT)
720-1335-6-ND - Digi-Reel®

Manufacturer [Vishay Vitramon](#)

Manufacturer Product Number VJ0603D471KXXAJ

Description CAP CER 470PF 25V C0G/NP0 0603

Manufacturer Standard Lead Time 15 Weeks

Customer Reference

Detailed Description

470 pF ±10% 25V Ceramic Capacitor C0G, NP0 0603 (1608 Metric)

Datasheet

[Datasheet](#)

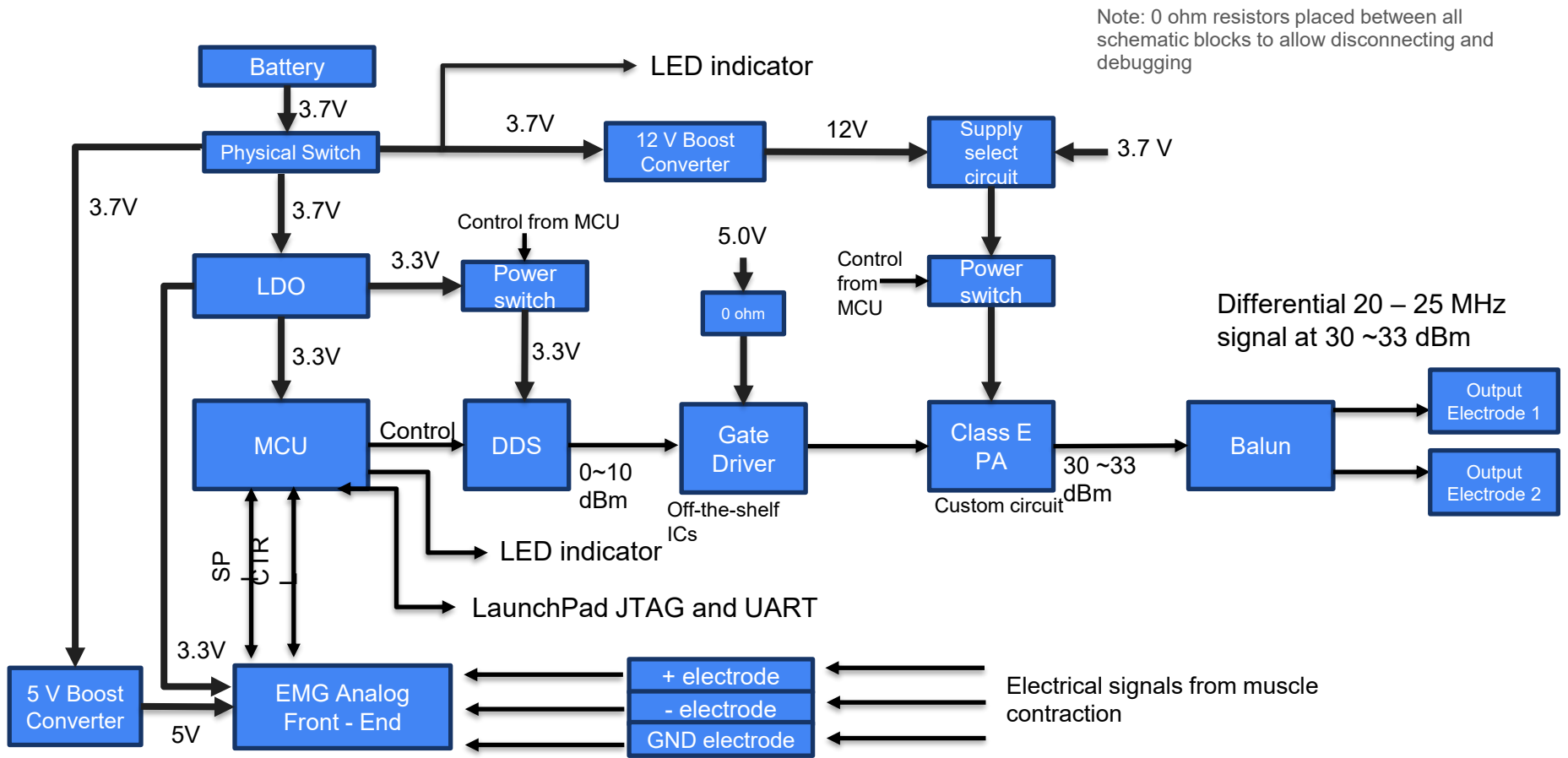
EDA/CAD Models

[VJ0603D471KXXAJ Models](#)

Observation/Conclusion

- We need to weigh the cost of troubleshooting this design versus dropping it in favor of the Class-A PA design that already has results.
- If we do continue trying to get a result out of the Class-E PA we may need to buy new parts (capacitors and inductors) with sufficient Q factors at the frequencies we plan to use the device in to avoid the large loss in power we experienced in our initial tests.
- We double check current rating of SMDs used to make sure they can handle about 1A to be safe.
- We can modify the design to increase the voltage at drain to compensate for lower current rating of SMDs.

Fabricated system architecture



Testing Class-E PA

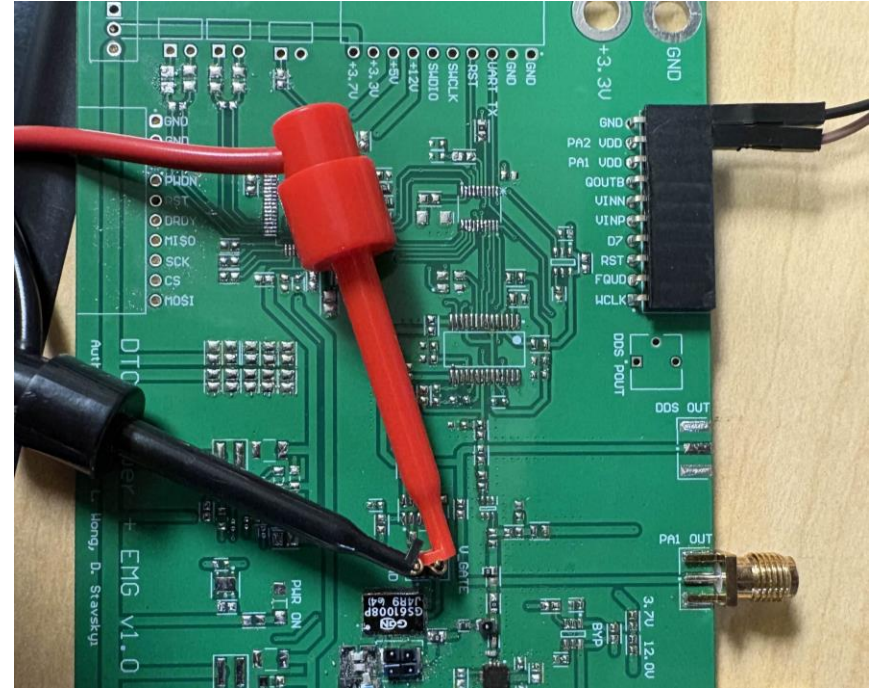
Luis Wong 01/28/2026

Objective

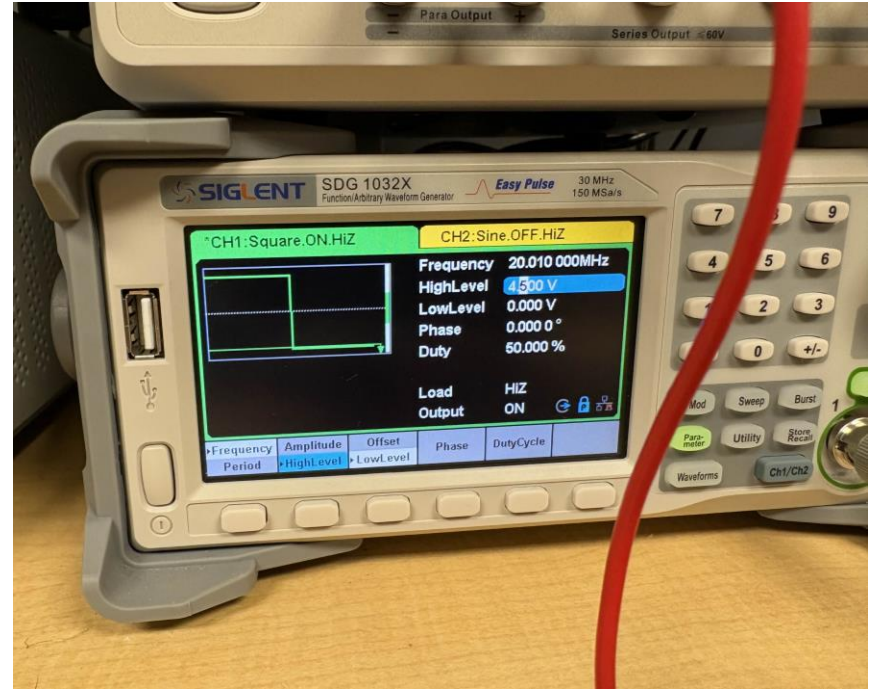
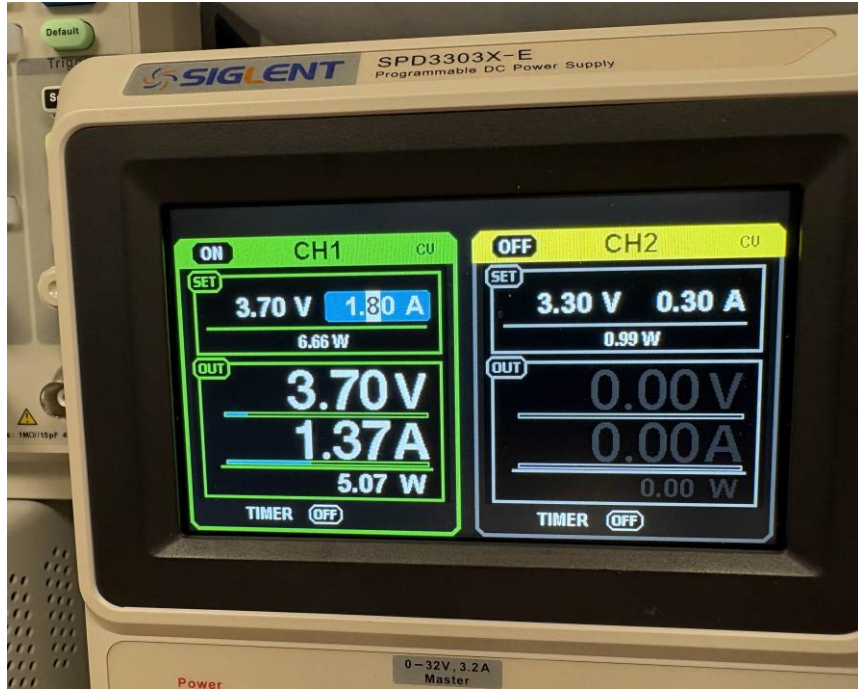
- Sweep frequencies of Class-E PA to check which frequencies the current design on the board works best with.

Methods

- Use spectrum analyzer, PSU, signal generator, and assembled PCB to test output power at different frequencies.

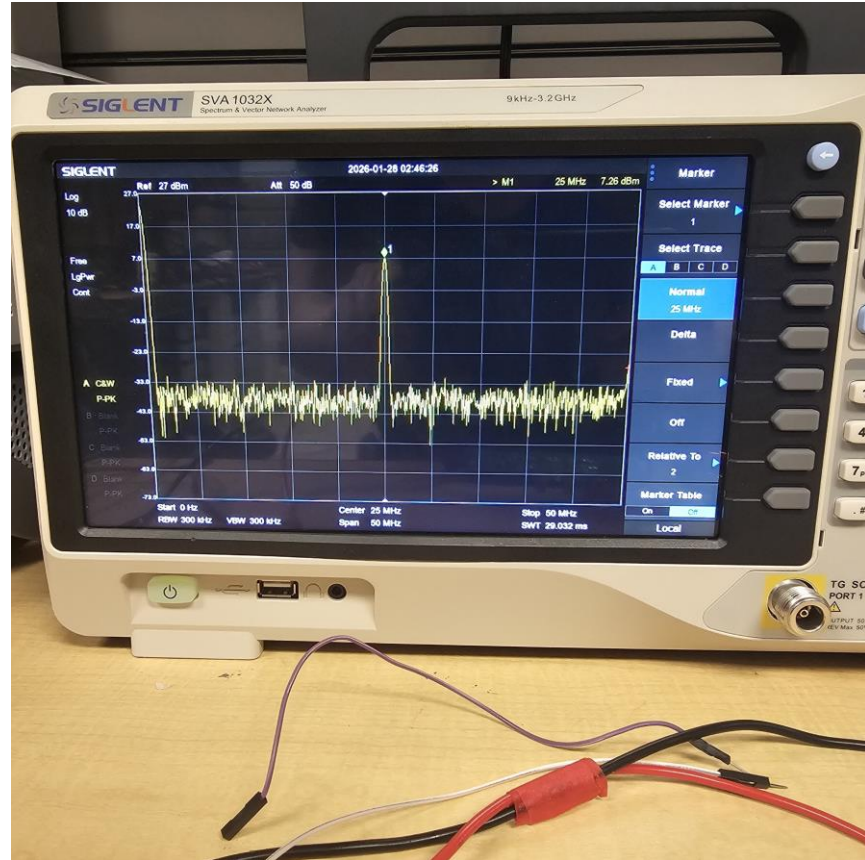


Results



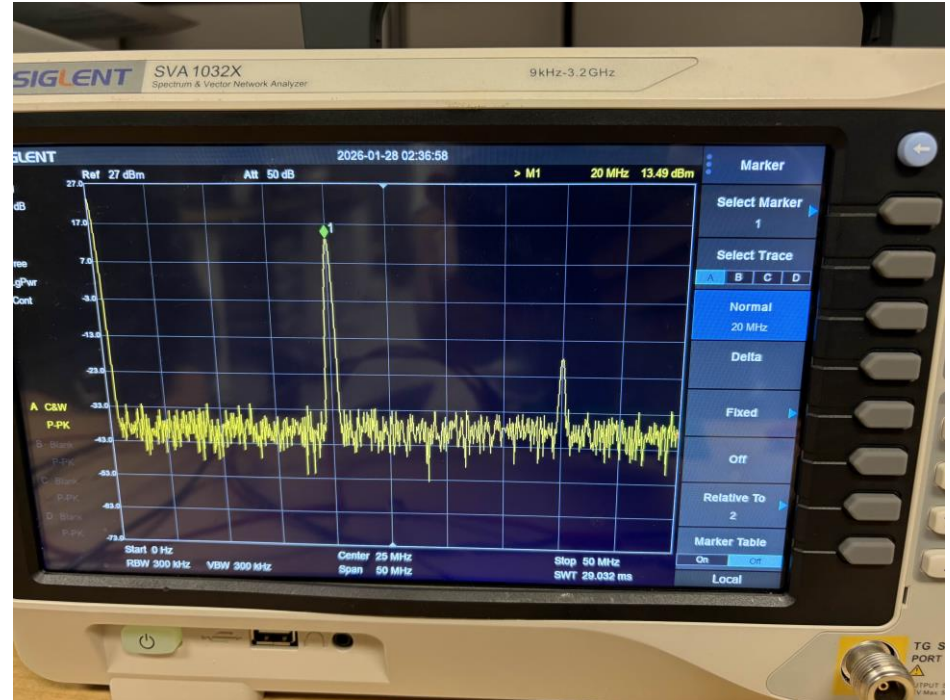
Results

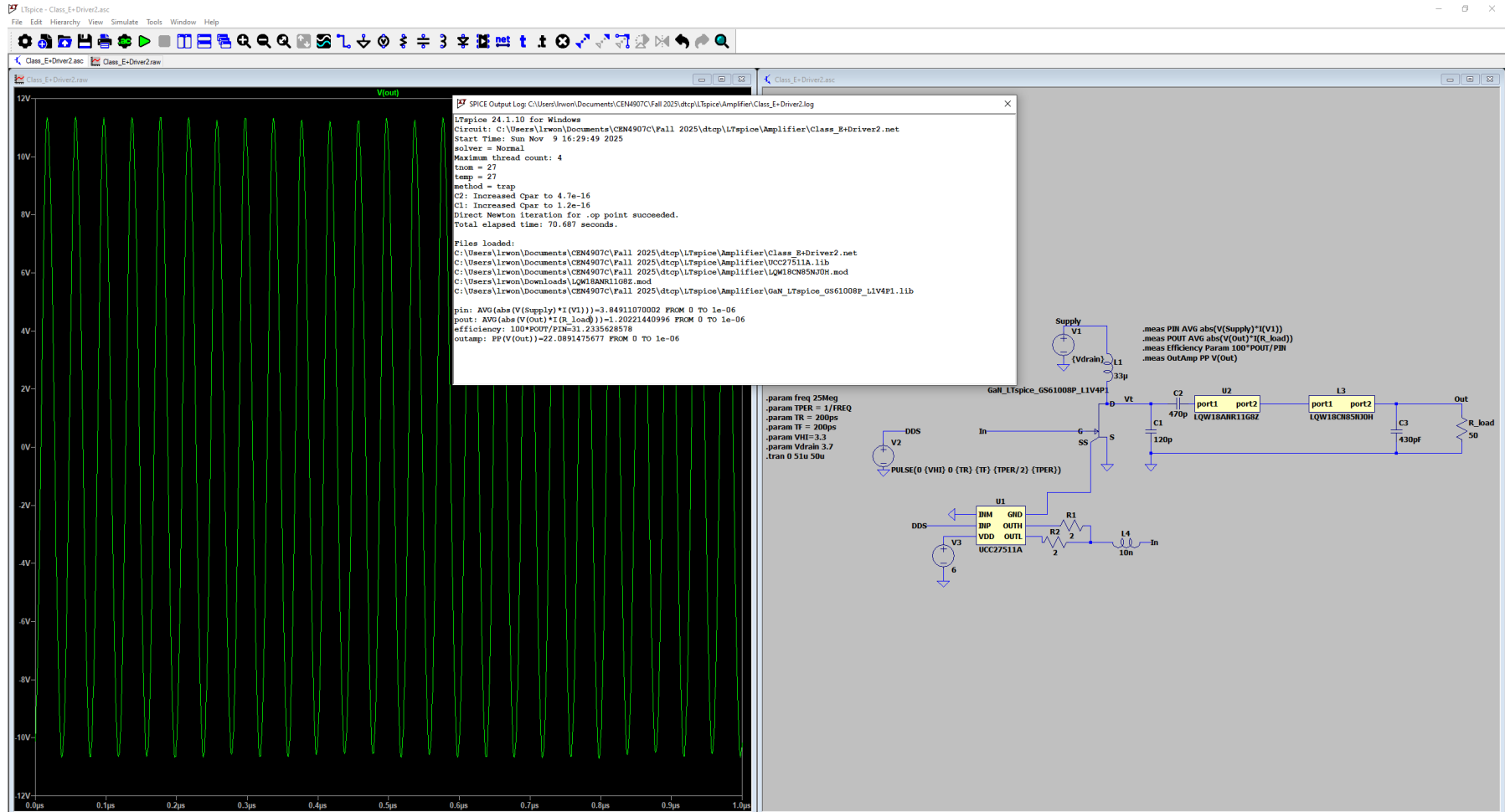
- At drain we have the same 1.5 μH inductor as before made use of four total with 2 pairs of them in series then connected in parallel.
- At 25 MHz after this change we are now getting 7.26 dBm instead of the negative 40 dBm we got before using a 0-4.5V square wave.
- When the signal at the gate is modified to a 0 to 4 V square wave with 3.3V or 5V at drain we get about negative 2 dBm. We seem to be getting the best results in terms of power output using a 0V - 4.5V square wave.



Results

- When sweeping frequencies, it becomes evident that the power output of the device peaks at around 18-20 MHz.
- Without changing the setup of using 3.7V at drain and a 0-4.5V square wave at the PA gave 13.5 dBm at 20 MHz.
- After increasing the drain voltage to 5V we saw that the signal amplified further to 15dBm but we had to stop there as we were worried we would damage the SMAs if we continued.





Observation/Conclusion

- With some modification to the design using 4 inductors at the drain instead of one we got better results. It seems we may just need to tune the rest of the circuit to get better results.
- Due to time constraints Jerimiah has opted to go with the two stage Class A PA as we already have it working and meeting specification requirements (30dBm).

Next Steps

- I have installed Code Composer Studio and tested its functionality with Dmytro.
- I will now be working on helping Dmytro with the Firmware and contributing to the second design with Jerimiah with Altium.

Simulating Class-E PA

Luis Wong 02/04/2026

Objective

- Simulate the design on the board using its parts and sweep frequencies.

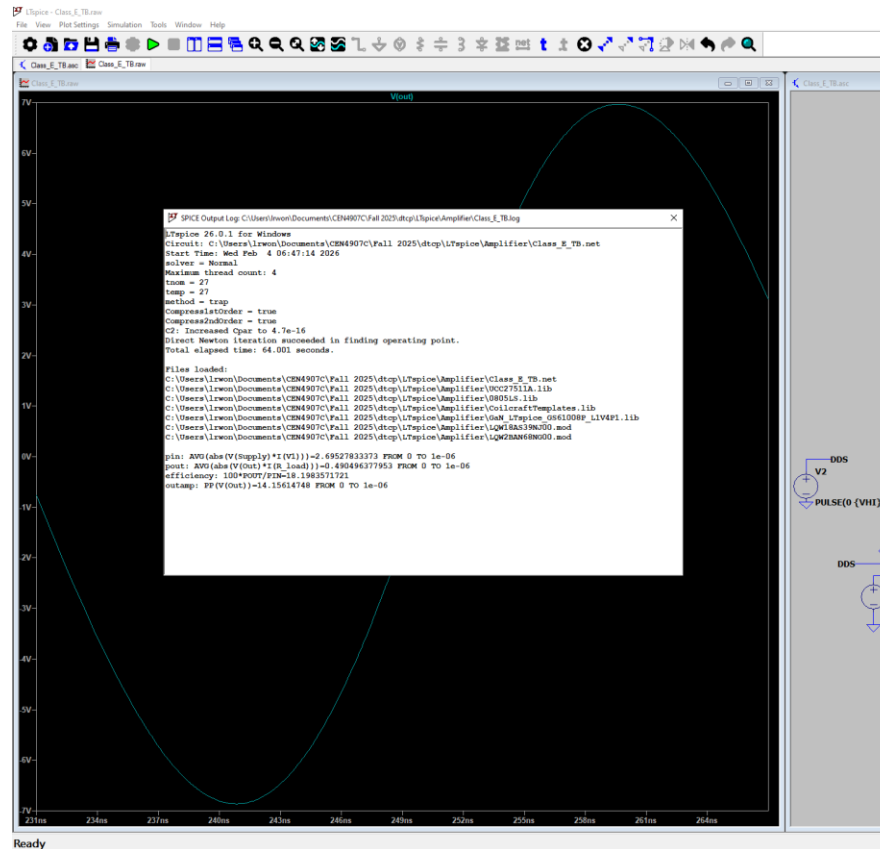
Methods

- LTspice

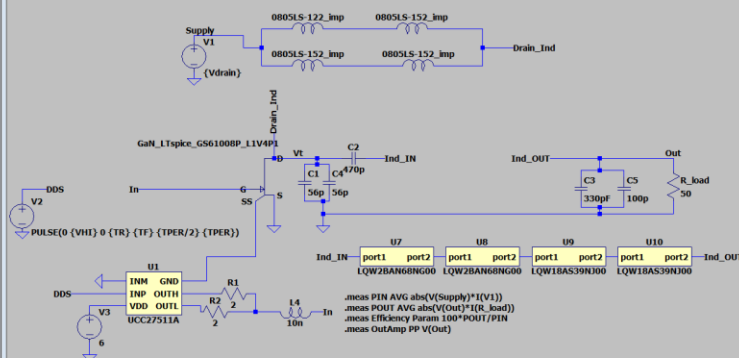
Target Specifications

Performance metric	Target value	Measured value as of 1/28/2026	Explanation
Output frequency	25 MHz	25 MHz	Measured with an RF spectrum analyzer.
Output power	≥ 30 dBm differential (1W)	7.26 dBm (0.0053 W)	Measured with an RF spectrum analyzer.
Static current	<10 mA	0 A (without power switches or driver)	Current consumed when the RF signal is not being transmitted. Class E PAs drive the transistor fully on and off like a switch. If the switch is off no DC path from supply to ground.
Tx current	<500 mA	1.37 A	Current consumed when the RF signal is being transmitted.

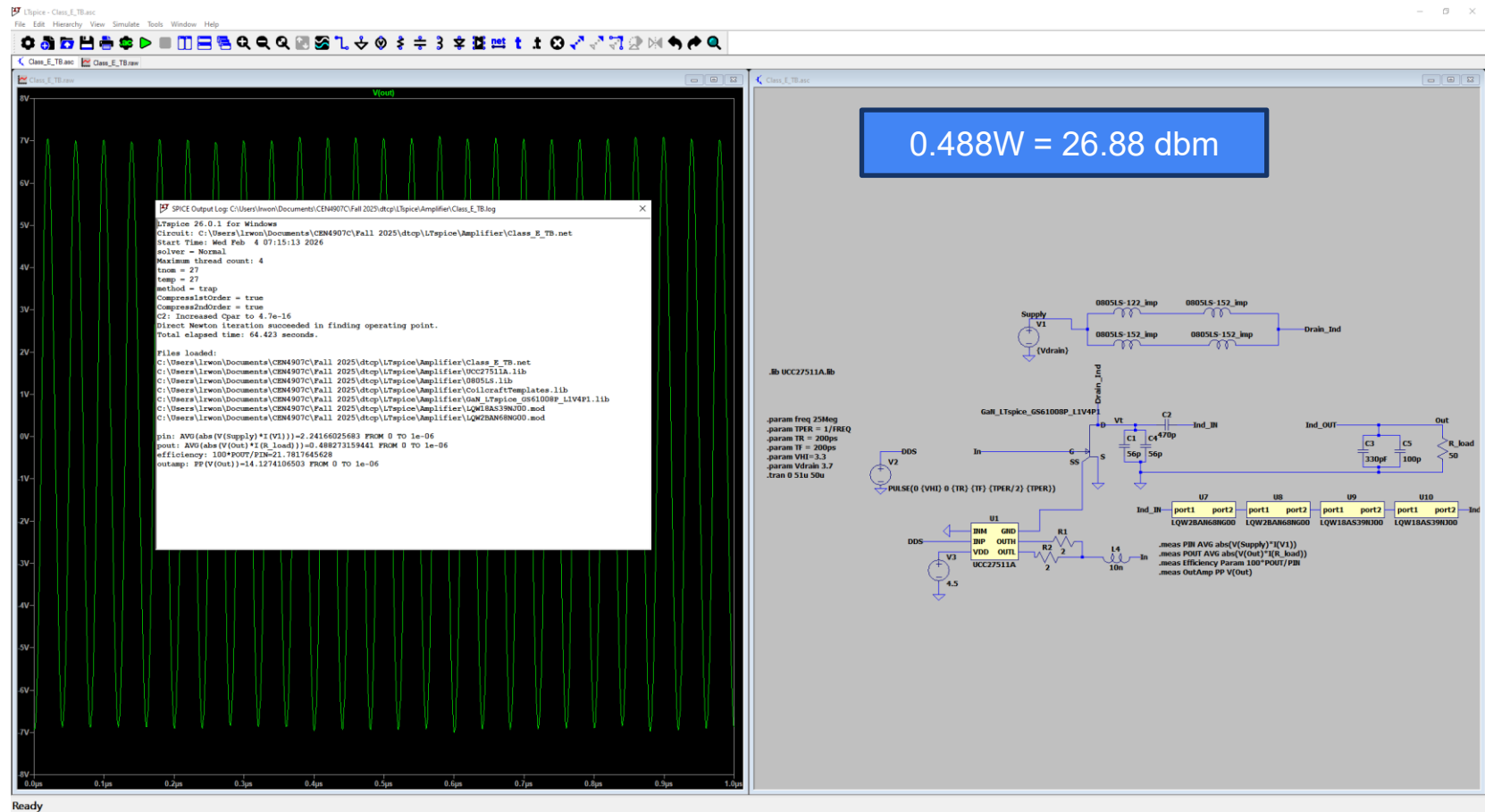
Results (6V square wave)



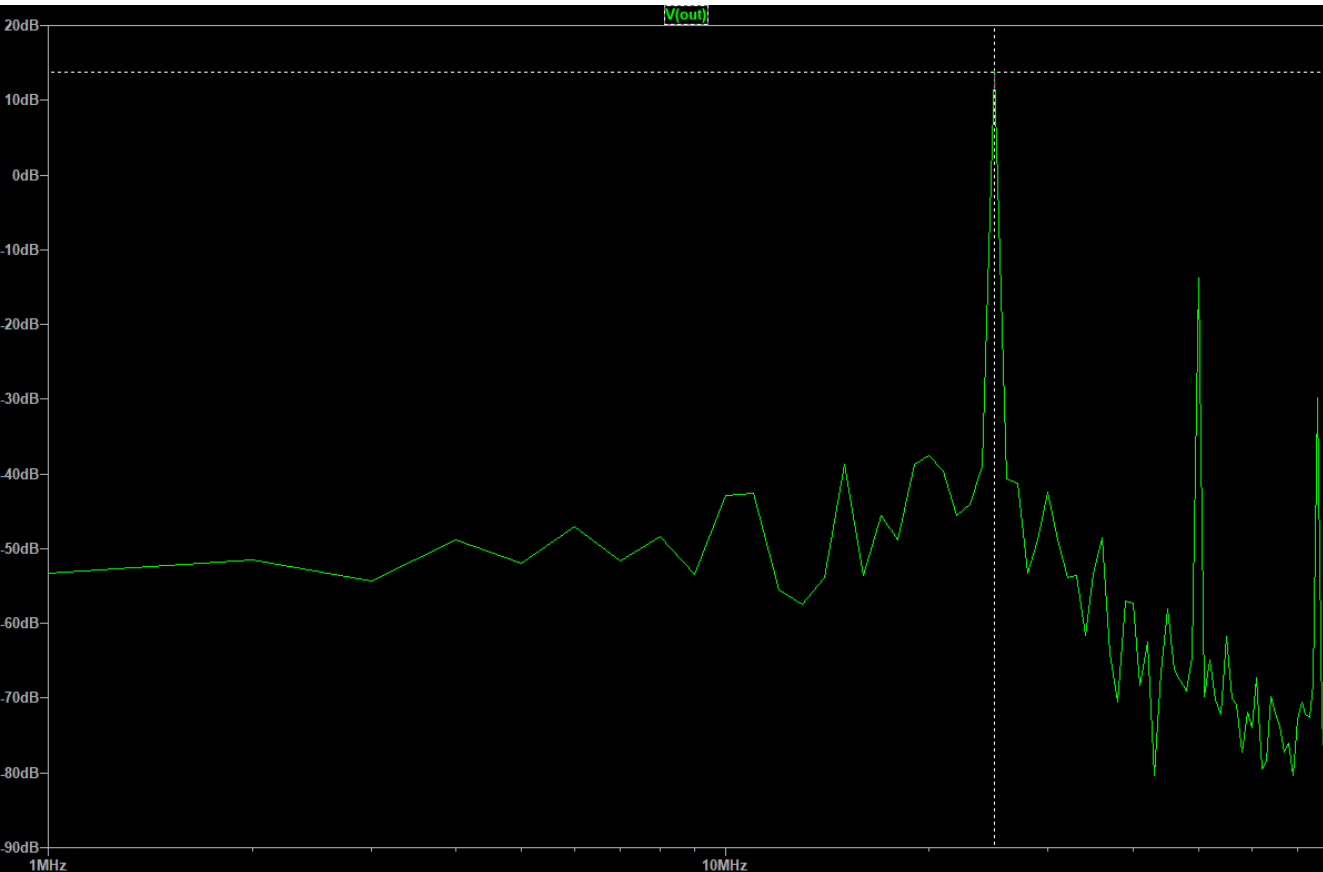
Power efficiency here is more accurately defined as drain efficiency.



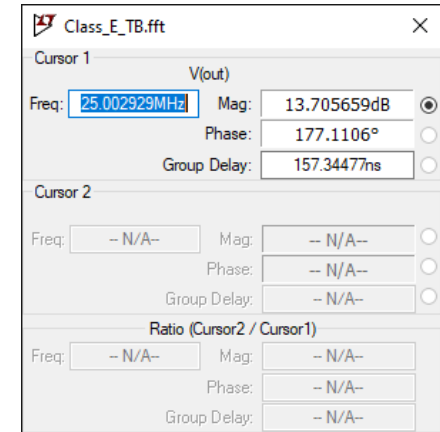
Results (4.5V square wave)

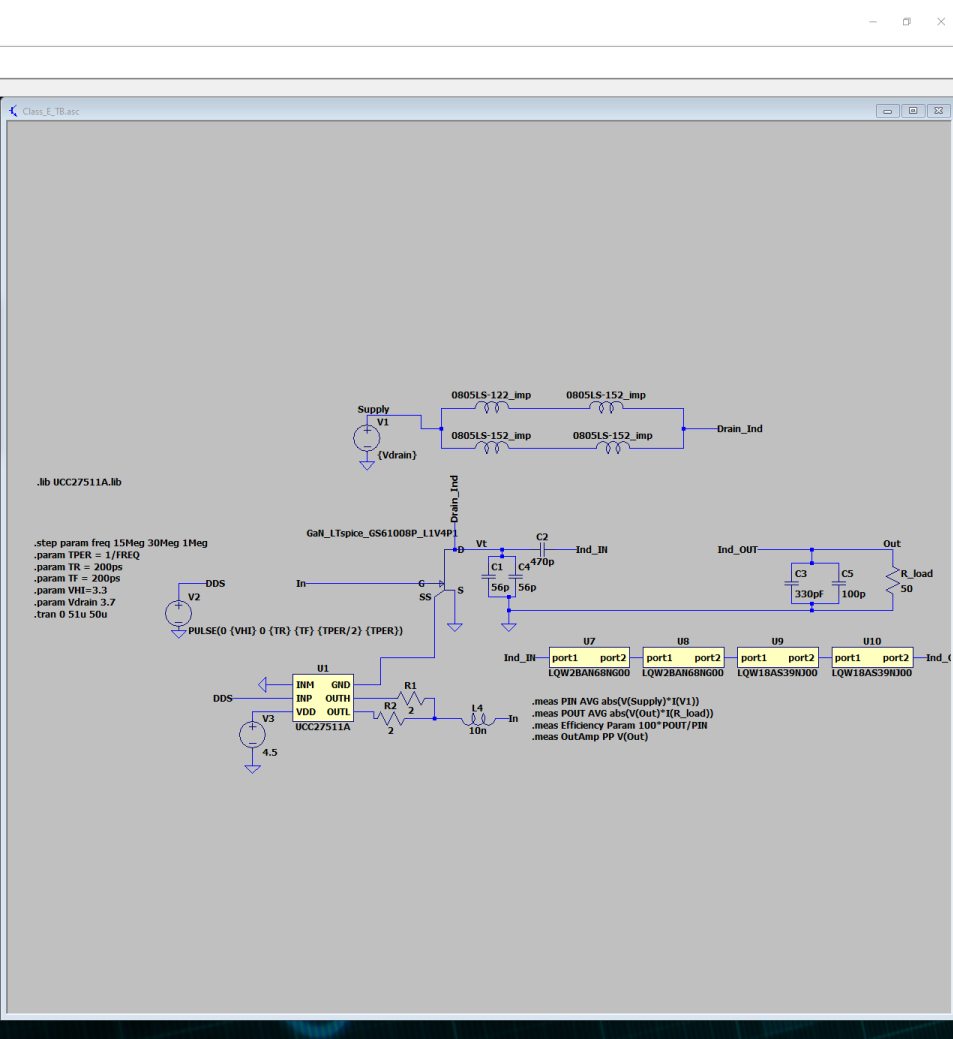
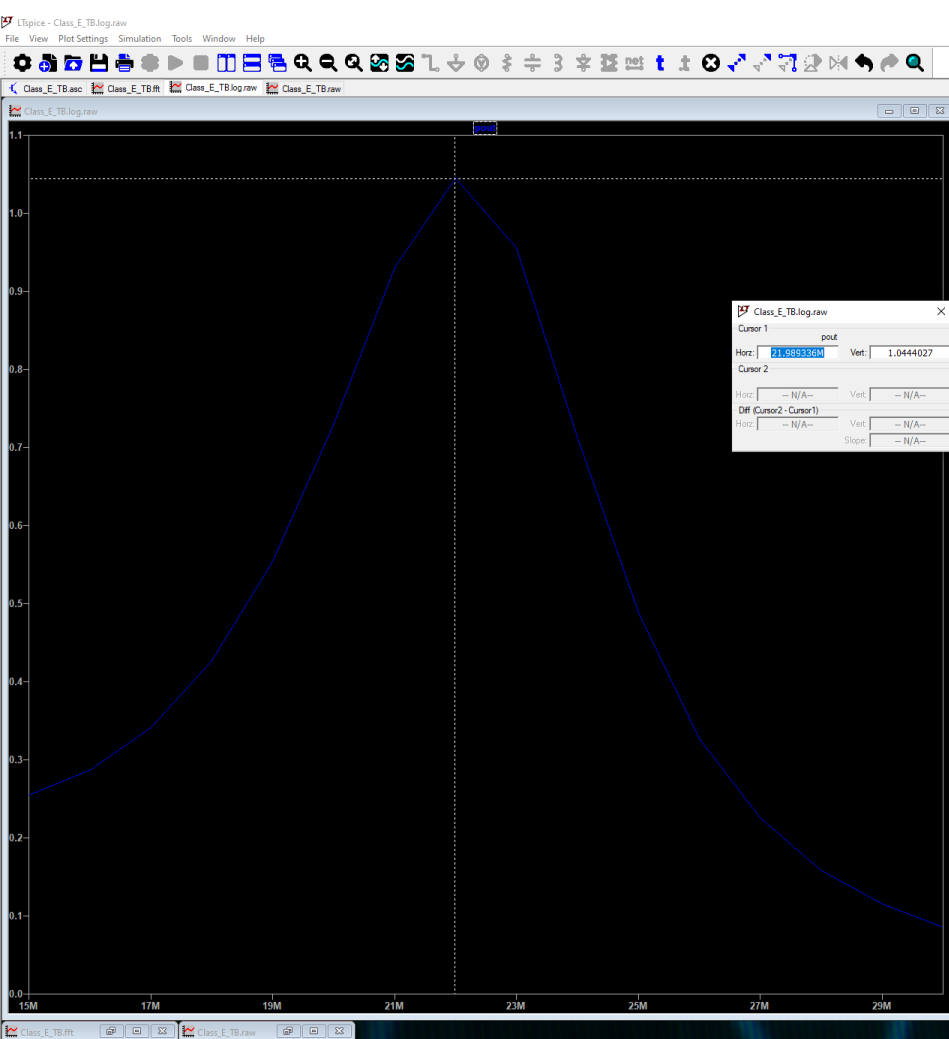


Results (4.5V square wave FFT)



Since I use transient analysis I calculated power using `.meas POUT AVG abs(V(Out)*I(R_load))`. This FFT done on the voltage confirms the signal is mostly 25 MHz.





Results (Calculating Theoretical Gain @ 25 MHz)

$$P_{in} = Q_G \cdot V_{GS} \cdot f$$

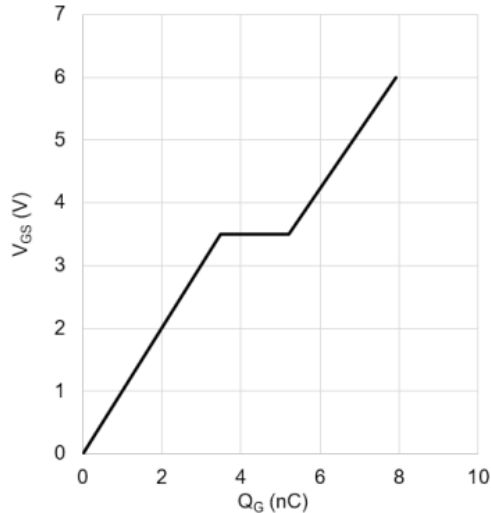
$$P_{in} = (6.3 \times 10^{-9} \text{ C}) \cdot (4.5 \text{ V}) \cdot (25 \times 10^6 \text{ Hz}) = 0.709 \text{ Watts}$$

$$\text{Gain} = \frac{P_{out}}{P_{in}}$$

$$\text{Gain} = \frac{0.488 \text{ W}}{0.709 \text{ W}} = 0.688$$

$$G_{db} = 10 * \log_{10} 0.688 = -0.162$$

Gate Charge, Q_G Characteristic

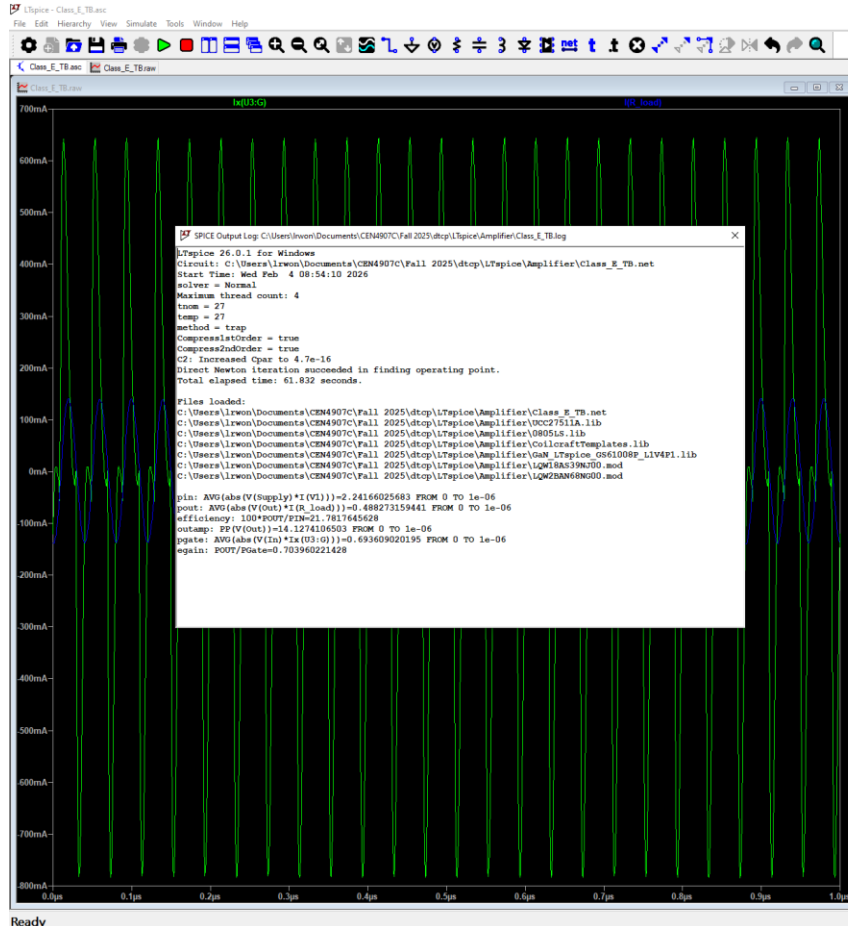


GS61008P
Bottom-side cooled 100 V E-mode GaN transistor
Datasheet

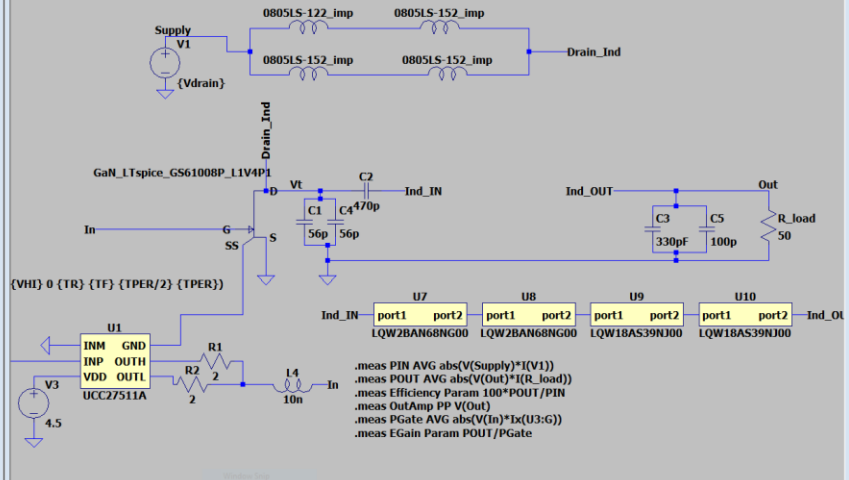
Electrical Characteristics (Typical values at $T_J = 25^\circ\text{C}$, $V_{GS} = 6 \text{ V}$ unless otherwise noted)

Parameters	Sym.	Min.	Typ.	Max.	Units	Conditions
Drain-to-Source Blocking Voltage	$V_{DS(BOSS)}$	100			V	$V_{GS} = 0 \text{ V}$, $I_{OSS} = 50 \mu\text{A}$
Drain-to-Source On Resistance	$R_{DS(on)}$		7	9.5	m Ω	$V_{GS} = 6 \text{ V}$, $T_J = 25^\circ\text{C}$, $I_{DS} = 27 \text{ A}$
Drain-to-Source On Resistance	$R_{DS(on)}$		17.5		m Ω	$V_{GS} = 6 \text{ V}$, $T_J = 150^\circ\text{C}$, $I_{DS} = 27 \text{ A}$
Gate-to-Source Threshold	$V_{GS(th)}$	1.1	1.7	2.6	V	$V_{DS} = V_{GS}$, $I_{DS} = 7 \text{ mA}$
Gate-to-Source Current	I_{GS}		200		μA	$V_{GS} = 6 \text{ V}$, $V_{DS} = 0 \text{ V}$
Gate Plateau Voltage	V_{plate}		3.5		V	$V_{DS} = 50 \text{ V}$, $I_{DS} = 90 \text{ A}$
Drain-to-Source Leakage Current	I_{DSS}		0.5	50	μA	$V_{DS} = 100 \text{ V}$, $V_{GS} = 0 \text{ V}$, $T_J = 25^\circ\text{C}$
Drain-to-Source Leakage Current	I_{DSS}		100		μA	$V_{DS} = 100 \text{ V}$, $V_{GS} = 0 \text{ V}$, $T_J = 150^\circ\text{C}$
Internal Gate Resistance	R_G		0.8		Ω	$f = 5 \text{ MHz}$, open drain
Input Capacitance	C_{iss}		600		pF	$V_{DS} = 50 \text{ V}$, $V_{GS} = 0 \text{ V}$, $f = 100 \text{ kHz}$
Output Capacitance	C_{oss}		250		pF	
Reverse Transfer Capacitance	C_{rss}		12		pF	
Effective Output Capacitance, Energy Related (Note 4)	$C_{O(ER)}$		302		pF	$V_{GS} = 0 \text{ V}$
Effective Output Capacitance, Time Related (Note 5)	$C_{O(TR)}$		385		pF	$V_{DS} = 0 \text{ to } 50 \text{ V}$
Total Gate Charge	Q_G		8		nC	$V_{GS} = 0 \text{ to } 6 \text{ V}$, $V_{DS} = 50 \text{ V}$, $I_{DS} = 90 \text{ A}$
Gate-to-Source Charge	Q_{GS}		3.5		nC	
Gate threshold charge	$Q_{G(th)}$		1.9		nC	
Gate switching charge	$Q_{G(sw)}$		3.3		nC	
Gate-to-Drain Charge	Q_{GD}		1.7		nC	
Output Charge	Q_{OSS}		20		nC	$V_{GS} = 0 \text{ V}$, $V_{DS} = 50 \text{ V}$
Reverse Recovery Charge	Q_{RR}		0		nC	

Results (LTspice Gain Calc)



Power at gate = 0.694
Gain = 0.704
Similar to calculated result.



Observation/Conclusion

- In the simulation it show that at 25MHz we should expect a theoretical power output of 26.88 dBm or 0.488W.
- Power output changes to be about 30 dBm or 1W when we change the frequency at the gate to 22 MHz.
- Gain seems to be low which may disqualify the Class-E PA as a viable alternative with the current design found on the board.

Next Steps

- To be determined by Jerimiah and Han.